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Boundary Limited Observer Systems

Empirical Equivalence, Interface Refinement, and the Limits of Latent Reconstruction

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Abstract. Two internal states can produce the same report. A deterministic measurement identifies a target quantity exactly when the quantity has the same value for all states that produce each report. The set of possible reports can leave the existence of an exact state decoder undecided when the admissible models allow hidden labels. For every nonempty set of reports, one model takes the reports themselves as its states. A second model attaches a hidden binary label to each state. Both models produce the same reports under every fixed rule for processing the output. Only the first model permits exact state reconstruction. In a model with six reports, increasing the number of hidden labels per report from one to eight raises the number of indistinguishable state pairs from zero to 168. A second measurement identifies a target if it separates every pair of states with the same report and different target values. This condition specifies what an additional measurement would need to establish about an observer's internal state.

Keywords: observer interface, reachable observations, latent reconstruction, identifiability, quotient, interface refinement.

1. Introduction

A person can give the same report on two occasions. A scientist may know which response the person gave while several internal states remain consistent with that response. The same problem arises for an observer that uses its own reports to infer its internal state. In either case, the question is which differences between states survive the measurement.

A known observation map permits a direct test of whether a quantity can be recovered. When only the set of possible outputs is known, recovery also depends on which source models are admissible. Statistical identifiability compares probability laws, and dynamical observability uses records over time (Rothenberg, 1971; Hermann and Krener, 1977). The evidence considered here consists of the possible outputs of a deterministic map. Model restrictions play a comparable role in the recovery of latent representations from data (Locatello et al., 2019).

For any nonempty set of possible outputs, two models illustrate how recovery depends on these restrictions. In one model the output identifies its source. In the other, two hidden labels for each source give identical outputs. Finite examples quantify how many states a report can conceal. A condition on a second measurement determines when the combined readings identify a specified target.

2. Observation and reconstruction

An interface is a map $\pi : X \to O$ from a source state space to an observation space. Two states are observationally equivalent when their outputs agree:

$$x \sim_\pi y \iff \pi(x) = \pi(y), \qquad Q_\pi = X/\!\sim_\pi. \tag{1}$$

Each class is a fiber of π. The quotient Q_π is in bijection with the reachable range $\pi[X]$. The observation determines a quantity $f : X \to V$ on this range exactly when f is constant on each fiber. In that case f factors uniquely through the quotient. Constructing an algorithm or bounding its error requires further assumptions.

A globally exact decoder is a total map $d : O \to X$ such that

$$d \circ \pi = \mathrm{id}_X . \tag{2}$$

Applying d to $\pi(x) = \pi(y)$ gives $x = y$, so exact decoding requires injectivity. For nonempty X, injectivity also suffices. Invert π on its range and choose one fixed state for unreachable observations. When $X = \varnothing$, a decoder defined on all of O exists exactly when $O = \varnothing$.

A model $\mathsf{M} = (X_\mathsf{M}, \pi_\mathsf{M})$ has reachable set $\mathscr{O}(\mathsf{M}) = \pi_\mathsf{M}[X_\mathsf{M}]$. Models with the same observation codomain are compared by

$$\mathsf{M} \approx \mathsf{M}' \iff \mathscr{O}(\mathsf{M}) = \mathscr{O}(\mathsf{M}'). \tag{3}$$

Empirical equivalence compares sets of possible observations. Their probabilities can differ. A fixed function $e : O \to R$ processes an observation into an experimental outcome. Its possible outcomes are

$$\mathscr{R}_e(\mathsf{M}) = e[\mathscr{O}(\mathsf{M})], \qquad \mathsf{M} \approx \mathsf{M}' \implies \mathscr{R}_e(\mathsf{M}) = \mathscr{R}_e(\mathsf{M}'). \tag{4}$$

The function e acts only on the existing observation. An intervention or a sequence of measurements requires a model of how the system evolves.

Interfaces on the same source space can retain different distinctions. A fine interface $\pi_F : X \to O_F$ refines $\pi_C : X \to O_C$ when a coarsening map c satisfies

$$\pi_C = c \circ \pi_F . \tag{5}$$

This relation is reflexive and transitive. It induces a surjection $[x]_F \mapsto [x]_C$ between observable quotients. If a pair is separated by F and merged by C, that quotient map is noninjective and reverse refinement is impossible. The proofs are in Appendices B to D.

3. Hidden states with identical outputs

A hidden binary label can prevent state reconstruction while preserving every output. For a model with $X_\mathsf{M} \neq \varnothing$, define

$$X_{\mathsf{M}^*} = X_\mathsf{M} \times \{0,1\}, \qquad \pi_{\mathsf{M}^*}(x,b) = \pi_\mathsf{M}(x), \qquad p(x,b) = x. \tag{6}$$

The map p is surjective and preserves observations. Distinct Boolean labels over the same base state produce a collision.

Theorem 3.1 (Hidden extension and reconstruction)**.** *For every model with a nonempty latent space, the extension in* (6) *has the same reachable set and the same possible outcomes under every fixed observation processing map e. It has no globally exact decoder. If the base has an exact decoder, the two models have different decoder statuses.*

Every base output $\pi_{\mathsf{M}}(x)$ occurs at $(x,0)$. Every extended output has this form. The two states $(x,0)$ and $(x,1)$ thus have the same observation. Equation (2) rules out an exact decoder for the extension (Appendices G and I).

This ambiguity holds for any nonempty set of possible observations $S \subseteq O$, provided the model class admits both constructions. Take the latent space to be S itself and the observation map to be the inclusion $S \hookrightarrow O$. An exact decoder returns the observed value on S and a fixed element of S elsewhere. The Boolean extension also reaches exactly S, but it has no exact decoder. For $S = \varnothing$ every compatible latent space is empty, and a total decoder exists exactly when $O = \varnothing$. Appendix H gives the proof.

Even all Boolean tests on observations leave this ambiguity. Such a test succeeds if some reachable output satisfies it. Tests for equality to individual outputs determine membership in the reachable set. Agreement on every test is thus equivalent to equality of reachable sets. A model property descends to the empirical quotient exactly when it is constant on each of these classes (Appendices F and H).

4. Methods

The finite calculations enumerate source states and group them by their outputs. The supplement gives the code, data tables, and Lean declarations for the corresponding mathematical results.

The finite base has $X = O = \{0,1,2,3,4,5\}$ and $\pi(x) = x$. A comparison model observes x mod 3. The implementation stores its three output labels. For comparison across models we include those labels in the common observation space O. Hidden extensions use $m \in \{1,2,3,5,8\}$ labels for each base state, with $\pi_m(x,h) = x$. For each reachable output, the program enumerates its fiber and calculates

$$C(\mathsf{M}) = \sum_{o \in \mathscr{O}(\mathsf{M})} \binom{|\pi_{\mathsf{M}}^{-1}(o)|}{2}, \qquad C(\mathsf{M}_m) = 3m(m-1). \tag{7}$$

Each pair contributes once when both states produce the same output.

Four rules process the output of each of seven representations of the six-output base. The rules return the original label, a threshold indicator, one of three categories, or the index of a bin of width two. Three predicates test evenness, values of at least three, and membership in $\{0,5\}$. A separate example compares four interfaces on eight states. The interfaces are identity, paired buckets, binary regions, and relabeled identity.

The program regenerates twelve CSV tables for comparison with the archived outputs. The supplement gives the commands to reproduce the calculations.

5. Results

All six outputs remain reachable as the number of hidden labels increases (Table 1). The injective base has six singleton fibers and an exact decoder. Its Boolean extension has twelve states and six

two-state fibers. Each fiber prevents exact recovery of its hidden label. The modulo three model has three reachable outputs and three indistinguishable pairs.

Table 1. Exhaustive hidden extension calculations. Each row reaches the same six observations.

m	Latent states	Reachable outputs	Fiber size	Collisions	Exact decoder
1	6	6	1	0	Yes
2	12	6	2	6	No
3	18	6	3	18	No
5	30	6	5	60	No
8	48	6	8	168	No

The four rules produce respectively six, two, three, and three possible outcomes. Each rule gives the same outcome set for all seven representations. Each of the three predicates also has the same truth table on the six reachable labels in all seven cases. These counts include the base itself and two ways of representing a two-label extension.

The identity, paired bucket, and binary region interfaces retain eight, four, and two classes, respectively. Grouping the identity outputs into pairs merges four previously distinct pairs of states. Grouping the paired outputs into regions merges eight more. Passing directly from identity to regions merges twelve pairs. Each map between these quotients is surjective and noninjective. Relabeling preserves all eight classes (Appendix L).

The 26 computational tests pass, and the twelve regenerated tables match the archived files byte for byte. The counts cover all states in the specified finite domains.

6. Discussion

A second measurement can distinguish states with the same report. Suppose the report is $\pi(x)$ and the target quantity is $f(x)$. A second reading $r(x)$ identifies the target from the joint observations exactly when

$$\pi(x) = \pi(y) \text{ and } f(x) \neq f(y) \quad \Longrightarrow \quad r(x) \neq r(y) \qquad (x, y \in X). \tag{8}$$

Each fiber of (π, r) is the intersection of a report fiber with a fiber of the second measurement. Equation (8) makes f constant on each intersection. Other aspects of the source state can vary within that intersection.

Two models can reach the same outputs with different probabilities. Bernoulli laws with parameters $1/10$ and $9/10$ both permit zero and one, but assign different probabilities to one. Comparing these models statistically requires the induced probability laws (Rothenberg, 1971).

Observations taken at different times can distinguish states that share one instantaneous output. For the known system

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = 0, \qquad y = x_1, \tag{9}$$

one output leaves x_2 undetermined, while $y(0)$ and $y(T)$, for $T \neq 0$, determine both initial coordinates. Measurement error in the reconstructed second coordinate is amplified as $|T|$ becomes small. The derivation and error bound are in Appendix J. This distinction is consistent with dynamical observability theory (Hermann and Krener, 1977).

Neural applications require the same separation between source state, measurement, and reconstruction. Gardner and colleagues recovered toroidal population geometry from simultaneous grid-cell recordings (Gardner et al., 2022). A spatial tuning map and a time series of joint activity retain different information. To infer attraction from reconstructed geometry one also needs a model of evolution and evidence that trajectories approach the proposed invariant set.

A study of conscious experience could combine a report with a physiological measurement. The additional reading would have to distinguish every pair of states with the same report and different target values. Applying this criterion requires a definition of the target and independent evidence for the measurement model. Feedback through self-observation and continuity across scales raise further questions (Appendices O and P).

7. Conclusion

A measurement identifies exactly those target quantities that agree throughout each observation fiber. Knowledge of all possible reports can still leave open whether the state that produced them is exactly reconstructible. The inclusion model and its Boolean extension exhibit both possibilities for the same nonempty set of reports.

For an embedded observer, an explanation of internal state must specify what the report measures. A further reading or a justified physical restriction must resolve any differences in the target that the report leaves open.

A. Mathematical preliminaries

A.1 Functions, images, and fibers

The notation for functions, images, fibers, composition, inverses, and equivalence relations follows standard set-theoretic usage (Halmos, 1974, Chs. 8–11). For a function $f : X \to Y$, the image of a subset $A \subseteq X$ is

$$f[A] = \{f(x) : x \in A\}.$$

The range is $\mathrm{im}(f) = f[X]$. The fiber over $y \in Y$ is

$$f^{-1}(y) = \{x \in X : f(x) = y\}.$$

The function is injective when $f(x) = f(x')$ implies $x = x'$. It is surjective when every $y \in Y$ has at least one preimage.

A function $g : Y \to X$ is a left inverse of f when

$$g \circ f = \mathrm{id}_X .$$

A left inverse can exist only when f is injective (Mac Lane and Birkhoff, 1999, Chs. 1 and 4).

A.2 Equivalence relations and quotients

An equivalence relation partitions a set into disjoint equivalence classes. The quotient $X/\!\sim$ is the set of those classes. The canonical class map is

$$q : X \to X/\!\sim, \qquad q(x) = [x].$$

A function on X descends to the quotient precisely when it is constant on each equivalence class. The descended function is then unique because every quotient class has a representative. This is the elementary set-level form of a quotient universal property (Mac Lane and Birkhoff, 1999; Leinster, 2014).

A.3 State-level and model-level notation

The symbol X denotes a source or world state space. The symbol O denotes an accessible observation space. The observation map is $\pi : X \to O$. A generative model is written

$$\mathsf{M} = (X_{\mathsf{M}}, \pi_{\mathsf{M}}).$$

Its reachable observation set is $\mathscr{O}(\mathsf{M})$. The possible outcome set of an admissible experiment e is $\mathscr{R}_e(\mathsf{M})$.

Observational equivalence of states identifies $x, y \in X$ when $\pi(x) = \pi(y)$. Empirical equivalence of models identifies M and M' when their reachable sets agree.

B. Observer interfaces, accessible quantities, and exact decoding

A single observation map determines which states produce the same output. In dynamical observability, the available evidence is an output trajectory generated by specified dynamics and inputs (Kalman, 1963; Hermann and Krener, 1977; Sontag, 1998).

Definition B.1 (Observer interface)**.** An observer interface on a world state space X is a pair

$$I = (O, \pi),$$

where O is an accessible observation space and $\pi : X \to O$ is the observation map.

Definition B.2 (Observational equivalence)**.** For $x, y \in X$, define

$$x \sim_I y \quad \Longleftrightarrow \quad \pi(x) = \pi(y).$$

Proposition B.3. *Observational equivalence is reflexive, symmetric, and transitive.*

Proof. Reflexivity follows from $\pi(x) = \pi(x)$. Symmetry follows because equality is symmetric. Transitivity follows because $\pi(x) = \pi(y)$ and $\pi(y) = \pi(z)$ imply $\pi(x) = \pi(z)$. □

The equivalence class of x is the fiber

$$[x]_I = \pi^{-1}(\pi(x)).$$

The class contains exactly the source states that the interface represents by the same accessible value. Equivalence is therefore relative to I. A second interface may separate states that the first interface merges.

B.1 Quantities that factor through the interface

An accessible quantity is a set function of the observation. Computability, cost, and stability under measurement error require additional hypotheses.

Definition B.4 (Interface accessible quantity)**.** Let $q : X \to V$. The quantity q factors through I when there exists a map $\bar{q} : O \to V$ such that

$$q = \bar{q} \circ \pi.$$

Theorem B.5 (Fiber constancy)**.** *If q factors through I, then q is constant on every observation fiber. In particular,*

$$x \sim_I y \quad \Longrightarrow \quad q(x) = q(y).$$

Proof. Assume $q = \bar{q} \circ \pi$. Let $x \sim_I y$. By definition, $\pi(x) = \pi(y)$. Applying $\bar{q}$ to both sides gives

$$\bar{q}(\pi(x)) = \bar{q}(\pi(y)).$$

Substitution of $q = \bar{q} \circ \pi$ yields $q(x) = q(y)$. □

A quantity constant on every fiber has a unique representation on the observable quotient (Appendix C).

Corollary B.6 (Variation prevents interface factorization)**.** *If $x \sim_I y$ and $q(x) \neq q(y)$, then q does not factor through I.*

Proof. If q factored through I, fiber constancy would imply $q(x) = q(y)$, contradicting the assumption. □

B.2 Internal tests

An internal test is a special case of an accessible quantity.

Definition B.7 (Internal test)**.** An internal binary test is a function

$$t : O \to \{\text{false}, \text{true}\}.$$

Proposition B.8 (No internal test separates one fiber)**.** *If $x \sim_I y$, then*

$$t(\pi(x)) = t(\pi(y))$$

for every internal test t.

Proof. Apply t to both sides of $\pi(x) = \pi(y)$. □

Every test on O receives the same input for states in one fiber. Changing its output space cannot distinguish these states.

B.3 Exact reconstruction

An exact decoder is a left inverse of the observation map (Halmos, 1974, Chs. 10–11).

Definition B.9 (Globally exact decoder)**.** A globally exact decoder for I is a function $d : O \to X$ such that

$$d \circ \pi = \mathrm{id}_X .$$

Theorem B.10 (Exact decoding implies injectivity)**.** *If a globally exact decoder exists, then π is injective.*

Proof. Assume $d \circ \pi = \mathrm{id}_X$. Let $x, y \in X$ and assume $\pi(x) = \pi(y)$. Applying d gives

$$d(\pi(x)) = d(\pi(y)).$$

The left inverse condition reduces this equality to $x = y$. Therefore π is injective. □

Corollary B.11 (Noninjectivity prevents exact decoding)**.** *If π is noninjective, then no globally exact decoder exists.*

Proof. The statement is the contrapositive of Theorem B.10. □

A many to one observation may still identify a particular target or identify a state within a restricted source class.

B.4 Finite capacity obstruction

For finite sets, counting the possible outputs can rule out injectivity.

Theorem B.12 (Finite pigeonhole obstruction)**.** *Let X and O be finite. If*

$$|O| < |X|,$$

then every observation map $\pi : X \to O$ is noninjective. Therefore there exist distinct $x, y \in X$ such that $x \sim_I y$, and no globally exact decoder exists.

Proof. Assume for contradiction that π is injective. An injective map from a finite set X into a finite set O implies $|X| \leq |O|$. This contradicts $|O| < |X|$. Hence π is noninjective. The collision and decoder conclusions follow from the definitions and the preceding corollary. □

C. The observable quotient and universal factorization

The quotient of X by observational equivalence retains exactly the distinctions available from the measurement. Each of its elements consists of the states giving one reachable output (Halmos, 1974, Chs. 8–11). A quantity factors through this quotient when its value is independent of the representative (Mac Lane and Birkhoff, 1999; Leinster, 2014).

Definition C.1 (Observable quotient)**.** The observable quotient of X relative to I is

$$Q_I = X/\!\sim_I.$$

The canonical class map is

$$q_I : X \to Q_I, \qquad q_I(x) = [x]_I.$$

The quotient discards the identities of states inside each fiber and retains the partition induced by observation. It is neither larger nor smaller than the available distinctions in the following sense.

Proposition C.2 (Quotient equality)**.** *For $x, y \in X$,*

$$q_I(x) = q_I(y) \quad \Longleftrightarrow \quad x \sim_I y.$$

Proof. Two quotient classes are equal exactly when their representatives are related by the defining equivalence relation. Here that relation is equality of observations. □

Let $\mathrm{im}(\pi) \subseteq O$ denote the reachable observation range. Each quotient class has one observation value, and each reachable observation value has one quotient class.

Theorem C.3 (Canonical equivalence with the observation range)**.** *The map*

$$\Phi : Q_I \to \mathrm{im}(\pi), \qquad \Phi([x]_I) = \pi(x),$$

is well defined and bijective.

Proof. Well definedness follows because representatives of one class have equal observations. Surjectivity follows from the definition of $\mathrm{im}(\pi)$. For injectivity, suppose $\Phi([x]_I) = \Phi([y]_I)$. Then $\pi(x) = \pi(y)$, so $x \sim_I y$ and $[x]_I = [y]_I$. □

The quotient and the reachable range are therefore equivalent descriptions of accessible resolution. The quotient emphasizes which source states have been identified. The range emphasizes which observation values remain reachable.

C.1 Universal property

The quotient is characterized by a universal property. Universal properties identify an object by the unique maps that factor through it rather than by a choice of representatives (Leinster, 2014).

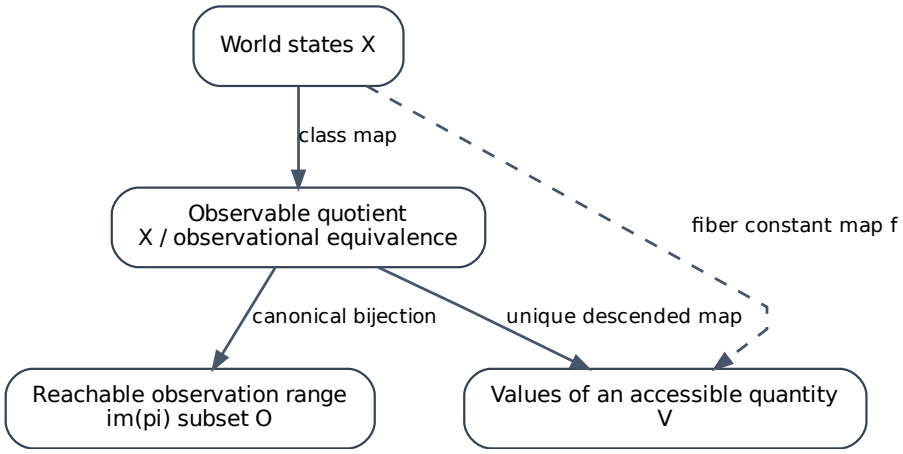


Figure 1. Universal factorization through the observable quotient. A quantity defined on world states descends uniquely to the quotient exactly when it is constant on observation fibers. The quotient is canonically equivalent to the reachable observation range.

Theorem C.4 (Observable quotient universal property)**.** *Let $f : X \to V$. The following statements are equivalent.*

(i) $f(x) = f(y)$ *whenever* $x \sim_I y$.

(ii) There exists a unique map $\bar{f} : Q_I \to V$ *such that*

$$f = \bar{f} \circ q_I.$$

Proof. Assume (i). Define $\bar{f}([x]_I) = f(x)$. The definition is independent of the representative because f is constant on each equivalence class. For every $x \in X$,

$$(\bar{f} \circ q_I)(x) = \bar{f}([x]_I) = f(x),$$

so the factorization exists. If $g : Q_I \to V$ also satisfies $f = g \circ q_I$, then for every class $[x]_I$,

$$g([x]_I) = g(q_I(x)) = f(x) = \bar{f}([x]_I).$$

Thus $g = \bar{f}$, proving uniqueness.

Conversely, assume (ii) and let $x \sim_I y$. Then $q_I(x) = q_I(y)$. Therefore

$$f(x) = \bar{f}(q_I(x)) = \bar{f}(q_I(y)) = f(y).$$

Hence f is constant on each fiber. □

The reachable observations determine a quantity precisely when it has a single value on each observation fiber.

C.2 Composition with the observation range

The original definition of interface factorization used a map $\hat{f} : O \to V$. The quotient theorem uses a map $\bar{f} : Q_I \to V$. These formulations agree on reachable observations. Extension to all of O is automatic when $V \neq \varnothing$, by choosing a value on unreachable observations. If $V = \varnothing$, a map $X \to V$ requires $X = \varnothing$; a map $O \to V$ then exists only if $O = \varnothing$. A function on the full codomain O may assign arbitrary values to unreachable observations, while a function on Q_I contains only empirically realized classes. The distinction becomes useful at the model level, where the empirical quotient is represented exactly by realizable reachable sets.

D. Interface refinement and information ordering

Instruments can measure the same states at different resolutions. Refinement defines a preorder on interfaces and an order on classes of mutually refining interfaces. Statistical experiment theory also compares informativeness (Blackwell, 1951, 1953; Torgersen, 1991). For deterministic observation maps, a coarsening function specifies the relation.

Let

$$F = (O_F, \pi_F) \qquad \text{and} \qquad C = (O_C, \pi_C)$$

be interfaces on the same world state space X.

Definition D.1 (Interface refinement)**.** The interface F refines C, written

$$F \succeq C,$$

when there exists a coarsening map $c : O_F \to O_C$ such that

$$\pi_C = c \circ \pi_F.$$

The coarsening function recovers every coarse observation from its corresponding fine observation. The reverse map may fail to exist.

Proposition D.2 (Refinement is reflexive and transitive)**.** *Every interface refines itself. If $F \succeq M$ and $M \succeq C$, then $F \succeq C$.*

Proof. Reflexivity uses the identity map on O_F. For transitivity, let $a : O_F \to O_M$ and $b : O_M \to O_C$ witness the two refinements. Then

$$\pi_C = b \circ \pi_M = b \circ a \circ \pi_F,$$

so $b \circ a$ witnesses $F \succeq C$. □

D.1 Monotonicity of accessible structure

A fine interface preserves every distinction visible to its coarsening.

Theorem D.3 (Fine equivalence implies coarse equivalence)**.** *If $F \succeq C$ and $x \sim_F y$, then $x \sim_C y$.*

Proof. Let c witness the refinement. The equality $\pi_F(x) = \pi_F(y)$ implies

$$c(\pi_F(x)) = c(\pi_F(y)).$$

Using $\pi_C = c \circ \pi_F$, this becomes $\pi_C(x) = \pi_C(y)$. □

The converse need not hold. A coarse interface may identify states that a fine interface distinguishes.

Theorem D.4 (Factorization is monotone under refinement)**.** *If $F \succeq C$ and $q : X \to V$ factors through C, then q factors through F.*

Proof. Let $q = \bar{q}_C \circ \pi_C$ and $\pi_C = c \circ \pi_F$. Then

$$q = \bar{q}_C \circ c \circ \pi_F .$$

Thus q factors through F by the map $\bar{q}_F = \bar{q}_C \circ c$. □

Every coarse test can therefore be simulated from the fine observation. Exact reconstruction also moves in the same direction.

Proposition D.5 (A coarse decoder induces a fine decoder)**.** *If $F \succeq C$ and C has a globally exact decoder, then F has a globally exact decoder.*

Proof. Let $d_C : O_C \to X$ satisfy $d_C \circ \pi_C = \mathrm{id}_X$, and let $c : O_F \to O_C$ witness refinement. Define $d_F = d_C \circ c$. Then

$$d_F \circ \pi_F = d_C \circ c \circ \pi_F = d_C \circ \pi_C = \mathrm{id}_X .$$

□

If the fine interface maps two states to the same output, every coarsening also maps them to the same output.

D.2 Refinement maps between observable quotients

Refinement induces a map between observable quotients.

Definition D.6 (Induced quotient map)**.** If $F \succeq C$, define

$$Q_{F \to C} : X/\!\sim_F \longrightarrow X/\!\sim_C$$

by

$$Q_{F \to C}([x]_F) = [x]_C .$$

The map is well defined because fine equivalence implies coarse equivalence.

Theorem D.7 (Surjectivity of the quotient map)**.** *For every refinement $F \succeq C$, the map $Q_{F \to C}$ is surjective.*

Proof. Let $[x]_C$ be any coarse class. The fine class $[x]_F$ maps to it by definition:

$$Q_{F \to C}([x]_F) = [x]_C .$$

Therefore every coarse class has a fine preimage. □

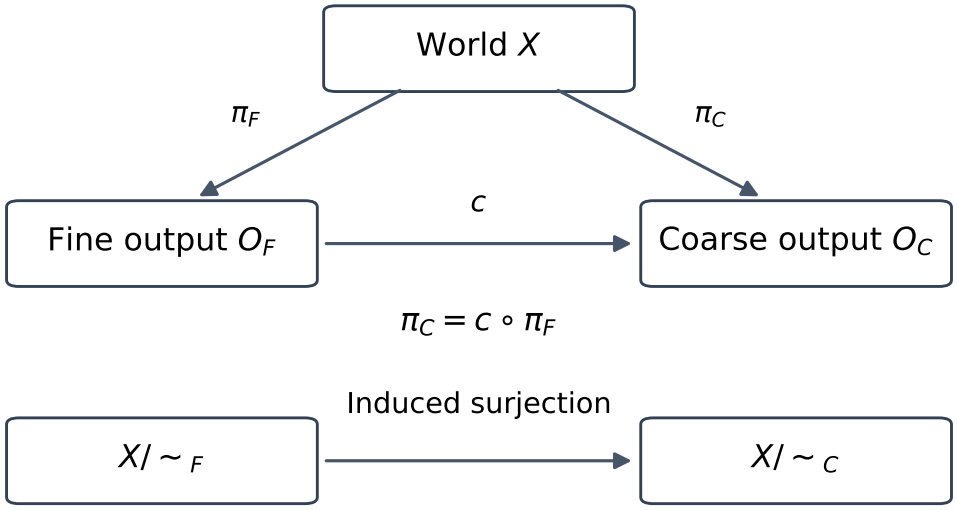


Figure 2. Interface refinement and induced quotient map. The coarsening map makes the observation triangle commute. The induced map sends each fine observational class to its coarse class. The map is always surjective. Under strict refinement it is not injective.

Definition D.8 (Strict refinement)**.** The refinement $F \succeq C$ is strict when there exist $x, y \in X$ such that

$$\pi_F(x) \neq \pi_F(y) \qquad \text{and} \qquad \pi_C(x) = \pi_C(y).$$

A strict refinement preserves at least one distinction that the coarse interface removes.

Theorem D.9 (Strict refinement makes the quotient map noninjective)**.** *If F strictly refines C, then $Q_{F\to C}$ is not injective.*

Proof. Choose x, y witnessing strictness. Since $\pi_F(x) \neq \pi_F(y)$, the fine classes are distinct:

$$[x]_F \neq [y]_F.$$

Since $\pi_C(x) = \pi_C(y)$, the coarse classes are equal. Hence

$$Q_{F\to C}([x]_F) = [x]_C = [y]_C = Q_{F\to C}([y]_F),$$

so the quotient map is noninjective. □

Theorem D.10 (Strict refinement prevents reverse refinement)**.** *If F strictly refines C, then C does not refine F.*

Proof. Suppose that $C \succeq F$. Fine equivalence would then follow from coarse equivalence. The strictness witnesses satisfy $x \sim_C y$ but $x \not\sim_F y$, which is a contradiction. □

D.3 Equal observational resolution

Two interfaces can use different observation labels while preserving exactly the same partition of X.

Definition D.11 (Same observational resolution)**.** Interfaces I and J have the same observational resolution when

$$x \sim_I y \quad \Longleftrightarrow \quad x \sim_J y$$

for every $x, y \in X$.

Mutual refinement implies equal observational resolution. Conversely, on reachable observations, equal resolution supplies mutually compatible class maps. The induced quotient maps are bijections.

Theorem D.12 (Mutual refinement gives quotient equivalence)**.** *If $I \succeq J$ and $J \succeq I$, then the induced quotient maps are inverse bijections. Thus the two interfaces retain the same observational distinctions even when their observation representations differ.*

This distinction separates strict coarsening from relabeling. Strict coarsening changes the partition of the world. Relabeling changes the symbols used for the classes but preserves the classes themselves.

E. Generative models and admissible experiments

The state level framework extends to models with different latent spaces. The common comparison object is the accessible observation space. Comparing models by the outputs they can generate is adjacent to structural identifiability and statistical experiment theory, although the present definition uses equality of reachable sets rather than equality of distributions or parameterized trajectories (Rothenberg, 1971; Walter and Pronzato, 1997; Blackwell, 1951; Torgersen, 1991).

Definition E.1 (Generative model)**.** A generative model over an observation space O is a pair

$$\mathsf{M} = (X_{\mathsf{M}}, \pi_{\mathsf{M}}),$$

where X_{M} is the latent state space and $\pi_{\mathsf{M}} : X_{\mathsf{M}} \to O$ is the observation map.

Definition E.2 (Reachable observation set)**.** The reachable observation set of M is

$$\mathscr{O}(\mathsf{M}) = \{\pi_{\mathsf{M}}(x) : x \in X_{\mathsf{M}}\} = \mathrm{im}(\pi_{\mathsf{M}}).$$

Definition E.3 (Empirical equivalence)**.** Two models M and M' are empirically equivalent when

$$\mathsf{M} \approx \mathsf{M}' \quad \Longleftrightarrow \quad \mathscr{O}(\mathsf{M}) = \mathscr{O}(\mathsf{M}').$$

Empirical equivalence is reflexive, symmetric, and transitive because set equality has those properties. The relation compares what the models can produce at the specified observation boundary. It does not compare latent identity directly.

E.1 Observation only experiments

An admissible experiment processes an existing observation. This follows the general methodological separation between an unobserved model state and the experiment or decision rule applied to its observable output (Blackwell, 1953; Le Cam, 1986; Le Cam and Yang, 2000).

Definition E.4 (Admissible experiment)**.** An admissible experiment is a function

$$e : O \to R.$$

Its input is an accessible observation. It does not receive a latent state.

Definition E.5 (Possible outcome set)**.** The possible outcome set of e under M is

$$\mathscr{R}_e(\mathsf{M}) = \{e(\pi_{\mathsf{M}}(x)) : x \in X_{\mathsf{M}}\} = e[\mathscr{O}(\mathsf{M})].$$

Theorem E.6 (Empirical equivalence preserves all admissible outcomes)**.** *If* $\mathsf{M} \approx \mathsf{M}'$*, then*

$$\mathscr{R}_e(\mathsf{M}) = \mathscr{R}_e(\mathsf{M}')$$

for every admissible experiment $e : O \to R$*.*

Proof. Empirical equivalence gives $\mathscr{O}(\mathsf{M}) = \mathscr{O}(\mathsf{M}')$. Applying the same function e to equal sets gives

$$e[\mathscr{O}(\mathsf{M})] = e[\mathscr{O}(\mathsf{M}')].$$

These images are the two possible outcome sets. □

The theorem applies to arbitrary result types. A Boolean test, categorical classifier, numerical compression, and identity map are all admissible experiments when their domain is O.

Definition E.7 (Experimental separation)**.** An experiment separates M and M' when

$$\mathscr{R}_e(\mathsf{M}) \neq \mathscr{R}_e(\mathsf{M}').$$

Corollary E.8 (No admissible separation inside one empirical class)**.** *If* $\mathsf{M} \approx \mathsf{M}'$*, no admissible experiment separates the models.*

E.2 The identity experiment

The identity experiment retains the complete accessible observation value.

Definition E.9 (Identity experiment)**.** Define

$$e_{\mathrm{id}} : O \to O, \qquad e_{\mathrm{id}}(o) = o.$$

Proposition E.10. *For every model* M*,*

$$\mathscr{R}_{e_{\mathrm{id}}}(\mathsf{M}) = \mathscr{O}(\mathsf{M}).$$

Proof. The image of $\mathscr{O}(\mathsf{M})$ under the identity function is $\mathscr{O}(\mathsf{M})$ itself. □

The identity experiment is useful because it rules out an explanation based only on downstream compression. Even when the experiment returns every reachable accessible observation unchanged, the observation set may still fail to determine latent cardinality, fiber structure, or exact reconstructability.

E.3 Properties determined by experiments

Let P be a property of generative models. The property is determined by a fixed experiment e when there exists a criterion C on possible outcome sets such that

$$P(\mathsf{M}) \quad \Longleftrightarrow \quad C(\mathscr{R}_e(\mathsf{M}))$$

for every model M. Any property determined in this way is invariant under empirical equivalence, because empirically equivalent models have equal outcome sets.

For $O \neq \varnothing$, the identity model and its Boolean extension have different decoder statuses inside one empirical class. Therefore it cannot be determined by any admissible experiment, including the identity experiment.

F. Predicate profiles and empirical identifiability

All Boolean predicates together identify the set of reachable outputs. Tests of external behavior also characterize equivalence in automata and process semantics (Moore, 1956; Park, 1981; Milner, 1989; Rutten, 2000). Here each predicate takes one observation as its argument.

Definition F.1 (Predicate success)**.** Let $p : O \to \{\text{false}, \text{true}\}$. The predicate p can succeed under M when

$$\exists x \in X_{\mathsf{M}} \quad p(\pi_{\mathsf{M}}(x)) = \text{true}.$$

Equivalently, at least one reachable observation satisfies p.

Definition F.2 (Predicate profile)**.** The predicate profile of M is the function

$$\mathsf{PredProf}(\mathsf{M}) : p \longmapsto \big[p \text{ can succeed under } \mathsf{M}\big].$$

Theorem F.3 (Empirical equivalence implies predicate equivalence)**.** *If $\mathsf{M} \approx \mathsf{M}'$, then every accessible predicate succeeds under M exactly when it succeeds under M'.*

Proof. Let p succeed under M. Then some $o \in \mathscr{O}(\mathsf{M})$ satisfies $p(o)$. Empirical equivalence gives $o \in \mathscr{O}(\mathsf{M}')$, so p succeeds under M'. The reverse direction is symmetric. □

The converse uses predicates that test equality to one observation.

Theorem F.4 (Complete predicate agreement implies empirical equivalence)**.** *Suppose every predicate $p : O \to \{\text{false}, \text{true}\}$ succeeds under M exactly when it succeeds under M'. Then*

$$\mathscr{O}(\mathsf{M}) = \mathscr{O}(\mathsf{M}').$$

Proof. Let $o \in \mathscr{O}(\mathsf{M})$. Define the equality predicate

$$p_o(a) = \begin{cases} \text{true}, & a = o, \\ \text{false}, & a \neq o. \end{cases}$$

The predicate succeeds under M, because o is reachable. By the assumed predicate agreement, it also succeeds under M'. Thus $o \in \mathscr{O}(\mathsf{M}')$. This proves $\mathscr{O}(\mathsf{M}) \subseteq \mathscr{O}(\mathsf{M}')$. The reverse inclusion follows by symmetry. □

Corollary F.5 (Predicate profile characterization)**.** *Two models are empirically equivalent if and only if their complete predicate profiles are equal.*

The complete profile therefore contains exactly the information in the reachable observation set. It is complete at the empirical boundary. It does not contain the multiplicities or identities of latent preimages.

F.1 Observation invariant and identifiable properties

Identifiability is always relative to a declared observation model and model class (Rothenberg, 1971; Walter and Pronzato, 1997; Raue et al., 2009; Villaverde et al., 2016). The definitions below isolate the corresponding invariance condition for the present reachable-set equivalence.

Definition F.6 (Observation invariant property)**.** A model property P is observation invariant when

$$\mathsf{M} \approx \mathsf{M}' \quad \Longrightarrow \quad \big(P(\mathsf{M}) \Longleftrightarrow P(\mathsf{M}')\big).$$

Definition F.7 (Empirically identifiable property)**.** A property is empirically identifiable from reachable outputs when its truth value is constant on every empirical equivalence class.

The two definitions are equivalent. The terminology emphasizes two views of the same condition. Observation invariance is a relation between representatives. Empirical identifiability is a property of the quotient class.

Any property determined by reachable observations, a fixed admissible experiment, or the complete predicate profile is observation invariant. The proof is the same in each case: empirically equivalent models supply equal criterion inputs.

Theorem F.8 (Exact latent reconstructability is not determined by any admissible experiment)**.** *Assume $O \neq \varnothing$ and a model class containing the identity model (O, id_O) and its Boolean hidden extension. For any fixed admissible experiment $e : O \to R$, no criterion on $\mathscr{R}_e(\mathsf{M})$ determines exact decoder existence for each model in that class.*

Proof. A property determined by an admissible experiment must be observation invariant. Choose $o_0 \in O$ and use the identity model, which has the identity decoder. In Appendix G, an exactly decodable model is paired with an empirically equivalent strict hidden extension that is not exactly decodable. Exact latent reconstructability is therefore not observation invariant. Hence no admissible experiment determines it. □

Corollary F.9. *Under the same assumptions, exact latent reconstructability is not determined by any accessible binary test, by the identity experiment, by the full reachable observation set, or by the complete predicate profile.*

This is stronger than a failure of one selected diagnostic. The complete collection of accessible predicates determines the empirical class, and the property still varies inside that class.

G. Hidden extensions and latent covers

The preceding sections identify what empirical equivalence preserves. Related identifiability results show that latent representations require additional assumptions to become unique (Teicher, 1963; Allman et al., 2009; Locatello et al., 2019; Hyvärinen et al., 2019; Khemakhem et al., 2020); the hidden extension constructs models with the same reachable observations and different latent spaces.

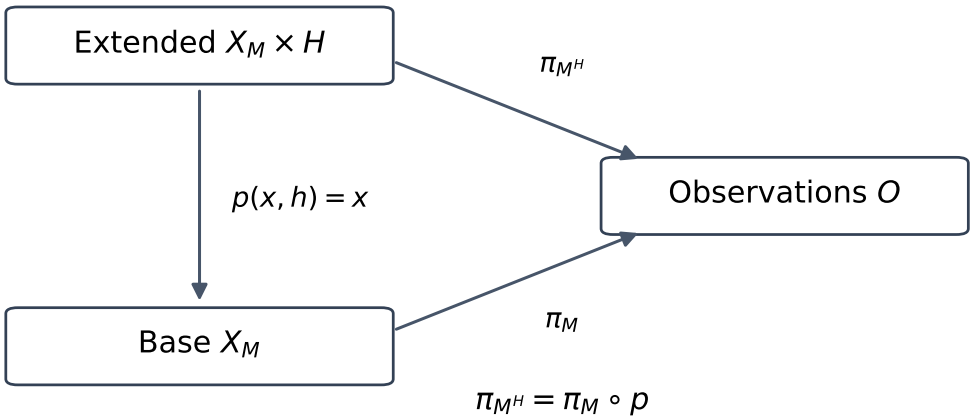


Figure 3. Hidden factor extension and observation preserving projection. The projection is surjective. Distinct hidden values attached to one base state create distinct extended states with one common observation.

G.1 Duplicate and hidden factor extensions

Let $\mathsf{M} = (X_{\mathsf{M}}, \pi_{\mathsf{M}})$ be a generative model.

Definition G.1 (Hidden factor extension)**.** For an inhabited set H, define the hidden factor extension M^H by

$$X_{\mathsf{M}^H} = X_{\mathsf{M}} \times H$$

and

$$\pi_{\mathsf{M}^H}(x,h) = \pi_{\mathsf{M}}(x).$$

The observation is independent of the hidden coordinate. Choosing a fixed anchor $h_0 \in H$ gives a surjective projection

$$p : X_{\mathsf{M}} \times H \to X_{\mathsf{M}}, \qquad p(x,h) = x.$$

The projection satisfies

$$\pi_{\mathsf{M}^H}(x,h) = \pi_{\mathsf{M}}(p(x,h)).$$

Theorem G.2 (Hidden extensions preserve empirical equivalence)**.** *For every inhabited H,*

$$\mathscr{O}(\mathsf{M}^H) = \mathscr{O}(\mathsf{M}).$$

Therefore $\mathsf{M}^H \approx \mathsf{M}$.

Proof. Every extended observation has the form $\pi_{\mathsf{M}^H}(x,h) = \pi_{\mathsf{M}}(x)$, so $\mathscr{O}(\mathsf{M}^H) \subseteq \mathscr{O}(\mathsf{M})$. Conversely, choose an anchor $h_0 \in H$. For every $x \in X_{\mathsf{M}}$, the extended state (x,h_0) produces $\pi_{\mathsf{M}}(x)$. Hence $\mathscr{O}(\mathsf{M}) \subseteq \mathscr{O}(\mathsf{M}^H)$. □

Corollary G.3 (Experimental invariance of hidden extensions)**.** *For every admissible experiment e,*

$$\mathscr{R}_e(\mathsf{M}^H) = \mathscr{R}_e(\mathsf{M}).$$

Theorem G.4 (Hidden collisions)**.** *If H contains distinct values $h_0 \neq h_1$ and X_{M} contains x, then*

$$(x, h_0) \neq (x, h_1)$$

but

$$\pi_{\mathsf{M}^H}(x, h_0) = \pi_{\mathsf{M}^H}(x, h_1).$$

Therefore the extended observation map is noninjective and M^H has no exact latent decoder.

Proof. The ordered pairs are distinct because their second coordinates differ. Both observations reduce by definition to $\pi_{\mathsf{M}}(x)$. The observation map is therefore noninjective. Exact latent decoding would imply injectivity, which is impossible. □

The Boolean extension uses $H = \{0, 1\}$. The duplicate model is the same two copy construction written as a specific finite hidden factor.

G.2 Finite hidden multiplicity

For a positive integer m, define

$$X_{\mathsf{M}^{(m)}} = X_{\mathsf{M}} \times \{0, 1, \ldots, m-1\}$$

and ignore the second coordinate during observation. Each base latent state generates m extended states in one observation fiber.

Proposition G.5 (Arbitrary finite hidden families)**.** *For every base state $x \in X_{\mathsf{M}}$, the map*

$$\iota_x : \{0, \ldots, m-1\} \to X_{\mathsf{M}^{(m)}}, \qquad \iota_x(j) = (x, j),$$

is injective, and every state in its image produces $\pi_{\mathsf{M}}(x)$.

Thus finite latent multiplicity can be increased without changing reachable observations or any admissible outcome set. In Lean, `finiteMultiplicityModel M n` uses the hidden type $\mathrm{Fin}(n+1)$; the label count m is therefore $n+1$.

G.3 Countable hidden multiplicity

The construction is not restricted to finite hidden sets. Let

$$X_{\mathsf{M}^{(\mathbb{N})}} = X_{\mathsf{M}} \times \mathbb{N}$$

and define

$$\pi_{\mathsf{M}^{(\mathbb{N})}}(x, n) = \pi_{\mathsf{M}}(x).$$

For each x, the map $n \mapsto (x, n)$ embeds $\mathbb{N}$ into the observation fiber over $\pi_{\mathsf{M}}(x)$. The model remains empirically equivalent to M, and every admissible outcome set remains equal. If $X_{\mathsf{M}} \neq \varnothing$, distinct natural number coordinates give a collision, so exact latent decoding is impossible.

This result separates empirical class size from latent fiber size. An empirical class can contain representatives with one, finitely many, or countably many latent states above the same accessible observation.

G.4 Latent covers

The product construction is one example of a more general relation.

Definition G.6 (Latent cover)**.** An extended model M^* is a latent cover of a base model M when there exists a surjective projection

$$p : X_{\mathsf{M}^*} \to X_{\mathsf{M}}$$

such that

$$\pi_{\mathsf{M}^*}(z) = \pi_{\mathsf{M}}(p(z))$$

for every $z \in X_{\mathsf{M}^*}$.

Latent covers compose. The identity map makes each model a cover of itself.

Theorem G.7 (A latent cover is empirically equivalent to its base)**.** *If* M^* *covers* M*, then*

$$\mathscr{O}(\mathsf{M}^*) = \mathscr{O}(\mathsf{M}).$$

Consequently, every admissible experiment has the same possible outcomes under both models.

Proof. Observation preservation shows that every observation of M^* is an observation of M. Surjectivity shows that every base latent state has an extended preimage, and observation preservation gives the same accessible output. Hence the reachable sets are equal. Outcome equality follows from Theorem E.6. □

Definition G.8 (Strict latent cover)**.** A latent cover is strict when its projection p is noninjective.

A strict cover contains distinct extended states that project to one base state. Observation preservation then forces those extended states to share one observation.

Theorem G.9 (Strict covers prevent exact latent reconstruction)**.** *If* M^* *is a strict latent cover of* M*, then* π_{M^*} *is noninjective and* M^* *has no exact latent decoder.*

Proof. Since the projection is noninjective, choose $z_1 \neq z_2$ with $p(z_1) = p(z_2)$. Observation preservation gives

$$\pi_{\mathsf{M}^*}(z_1) = \pi_{\mathsf{M}}(p(z_1)) = \pi_{\mathsf{M}}(p(z_2)) = \pi_{\mathsf{M}^*}(z_2).$$

Thus the extended observation map is noninjective. Exact decoding is therefore impossible. □

G.5 Existence of strict covers for inhabited models

Every inhabited model has a Boolean strict hidden extension. The construction can then be repeated.

Theorem G.10 (Every inhabited model admits a strict latent cover)**.** *For every generative model with at least one latent state, there exists an empirically equivalent strict latent cover with additional latent distinctions.*

Proof. Use the Boolean hidden extension. Projection onto the base state is surjective and observation preserving. The two Boolean values make the projection noninjective. □

The Lean predicate `LatentlyMaximal` means that no strict latent cover exists. The theorem negates that predicate for every inhabited model. This definition is not ordinary maximality in a partial order. A strict cover means a noninjective surjective projection; for infinite models such a projection may be a self-map. For example, $X = \mathbb{N}$, $O = \{0\}$, and $\pi(n) = 0$ admit the strict self-cover $p(n) = \lfloor n/2 \rfloor$. Surjectivity follows from $p(2n) = n$, and $p(0) = p(1)$ proves noninjectivity. Additional hidden labels also need not increase an infinite set's cardinality. The construction gives strict covers in the specified class. A physical interpretation requires independent evidence for the added variables.

H. The empirical model quotient

Empirical equivalence partitions generative models into classes with equal reachable observation sets. The empirical model quotient records only this accessible information. The descent results use the same representative-independence principle as ordinary quotient constructions and universal properties (Mac Lane and Birkhoff, 1999; Leinster, 2014), now applied to a space of models.

Definition H.1 (Empirical model quotient)**.** Let $\mathfrak{M}_O$ denote the collection of generative models over O, with a fixed set theoretic universe for latent spaces. The Lean development makes the corresponding universe parameters explicit. The empirical model quotient is

$$\mathfrak{M}_O/\!\approx,$$

where $\mathsf{M} \approx \mathsf{M}'$ means $\mathscr{O}(\mathsf{M}) = \mathscr{O}(\mathsf{M}')$.

The canonical projection sends each model to its empirical class:

$$q_{\mathscr{E}} : \mathfrak{M}_O \to \mathfrak{M}_O/\!\approx.$$

Two canonical classes are equal exactly when the corresponding models are empirically equivalent.

H.1 Invariant properties descend

Theorem H.2 (Universal property for model properties)**.** *A property $P : \mathfrak{M}_O \to \{\text{false}, \text{true}\}$ is observation invariant if and only if there exists a unique property*

$$\bar{P} : \mathfrak{M}_O/\!\approx \to \{\text{false}, \text{true}\}$$

such that

$$P = \bar{P} \circ q_{\mathscr{E}}.$$

Proof. If P is observation invariant, define $\bar{P}([\mathsf{M}]) = P(\mathsf{M})$. This is well defined because all representatives of one class have the same truth value. The factorization follows from the definition, and uniqueness follows because every quotient class has a representative. Conversely, any property factored through the quotient is constant on each quotient class and is therefore observation invariant. □

The same result holds for arbitrary valued measurements.

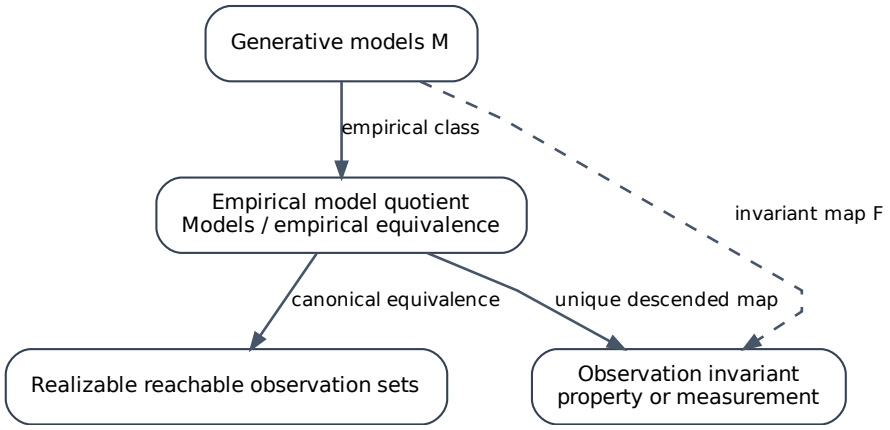


Figure 4. Empirical quotient and invariant descent. Observation invariant properties and measurements factor uniquely through empirical classes. Exact latent reconstructability fails this condition because it varies among representatives of one class.

Theorem H.3 (Universal property for invariant measurements)**.** *Let $F : \mathfrak{M}_O \to V$. The map F assigns equal values to empirically equivalent models if and only if there exists a unique*

$$\bar{F} : \mathfrak{M}_O/\approx \to V$$

such that $F = \bar{F} \circ q_{\mathscr{E}}$.

H.2 Realizable observation sets

The quotient has a concrete representation.

Definition H.4 (Realizable observation set)**.** A subset $A \subseteq O$ is realizable when there exists a generative model M such that

$$\mathscr{O}(\mathsf{M}) = A.$$

Theorem H.5 (Canonical representation of empirical classes)**.** *The map*

$$[\mathsf{M}] \longmapsto \mathscr{O}(\mathsf{M})$$

is a bijection between the empirical model quotient and the collection of realizable reachable observation sets.

Proof. The map is well defined because empirically equivalent models have equal reachable sets. It is injective because equal reachable sets imply empirical equivalence and therefore equal quotient classes. It is surjective by the definition of realizability. □

Each empirical class therefore specifies one reachable set. Whether a latent property is fixed within that class depends on the allowed models. For example, an empty reachable set forces an empty latent space.

H.3 Exact reconstruction does not descend

Theorem H.6 (Exact latent reconstruction is not a quotient property)**.** *Assume $O \neq \varnothing$ and a model class containing (O, id_O) and its Boolean hidden extension. Exact latent reconstructability does not factor through the empirical model quotient of that class.*

Proof. Use the identity model on O; it is inhabited and has the identity decoder. Its Boolean hidden extension is empirically equivalent to the base. The extension has a noninjective observation map and no exact latent decoder. Therefore the property has different truth values on two representatives of one empirical class. A property with that variation cannot descend to the quotient. □

The identity model gives a counterexample at the full observation space. An inclusion map gives a counterexample at any prescribed nonempty reachable set.

Corollary H.7 (Decoder status at a prescribed reachable set)**.** *Let $\varnothing \neq S \subseteq O$. If the model class contains the inclusion $i : S \hookrightarrow O$ and its Boolean extension, the empirical class for S contains both an exactly decodable model and one without a decoder. If the reachable set is empty, all compatible models have the same decoder status.*

Proof. Choose $s_0 \in S$. The map $d : O \to S$ that returns o for $o \in S$ and s_0 otherwise satisfies $d \circ i = \mathrm{id}_S$. Its Boolean extension reaches the same set, and $(s_0, 0)$ and $(s_0, 1)$ prevent exact decoding. If the reachable set is empty, a source state would give a reachable value, so the source is empty. A total map from O into the empty set exists precisely when O is empty. □

I. Synthesis theorem

Every model with a nonempty state space has a strict hidden extension that preserves all reachable outputs and prevents exact recovery of the extended state.

Theorem I.1 (Boundary limited observer synthesis)**.** *Let $\mathsf{M} = (X_\mathsf{M}, \pi_\mathsf{M})$ be a generative model, let $x_0 \in X_\mathsf{M}$ be an inhabited witness, and let $e : O \to R$ be an admissible experiment. Then there exists a model M^* such that:*

(i) M^* *is a strict latent cover of* M*;*
(ii) the empirical classes are equal, $q_{\mathscr{E}}(\mathsf{M}^*) = q_{\mathscr{E}}(\mathsf{M})$, equivalently $\mathsf{M}^* \approx \mathsf{M}$;
(iii) $\mathscr{O}(\mathsf{M}^*) = \mathscr{O}(\mathsf{M})$*;*
(iv) $\mathscr{R}_e(\mathsf{M}^*) = \mathscr{R}_e(\mathsf{M})$*; and*
(v) M^* *has no exact latent decoder.*

Proof. Choose the Boolean extension

$$X_{\mathsf{M}^*} = X_\mathsf{M} \times \{0, 1\}$$

and define

$$\pi_{\mathsf{M}^*}(x, b) = \pi_\mathsf{M}(x).$$

Define the projection

$$p : X_{\mathsf{M}^*} \to X_\mathsf{M}, \qquad p(x, b) = x.$$

The projection is surjective. For each $x \in X_\mathsf{M}$, the state $(x, 0)$ maps to x. It also preserves observation because

$$\pi_{\mathsf{M}^*}(x, b) = \pi_\mathsf{M}(p(x, b)).$$

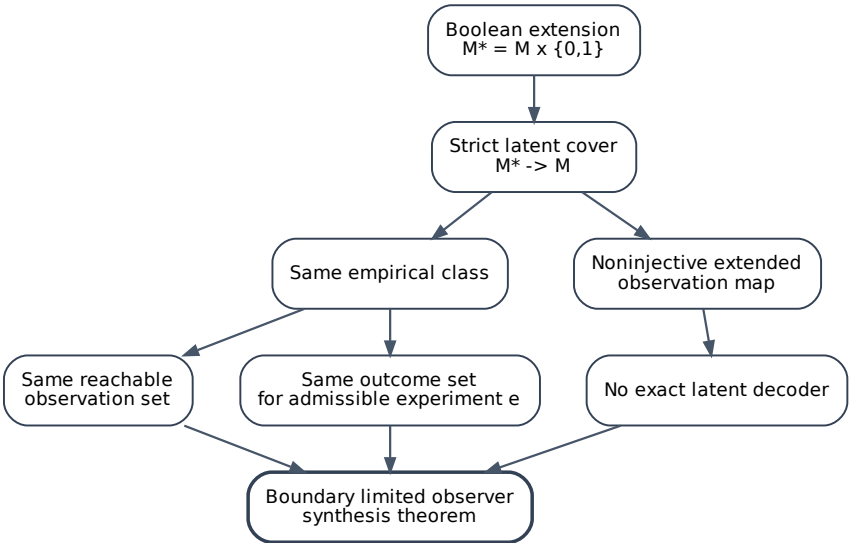


Figure 5. Dependency structure of the synthesis theorem. The result combines fiber based reconstruction limits, model outcome invariance, strict hidden extensions, and empirical quotient structure.

Thus M^* is a latent cover of M.

The cover is strict. Use the supplied witness $x_0 \in X_{\mathsf{M}}$. Then

$$(x_0, 0) \neq (x_0, 1)$$

but

$$p(x_0, 0) = x_0 = p(x_0, 1).$$

Hence the projection is noninjective.

Every latent cover is empirically equivalent to its base. Therefore

$$\mathsf{M}^* \approx \mathsf{M}$$

and

$$\mathscr{O}(\mathsf{M}^*) = \mathscr{O}(\mathsf{M}).$$

By empirical outcome invariance,

$$\mathscr{R}_e(\mathsf{M}^*) = \mathscr{R}_e(\mathsf{M}).$$

Finally, the two distinct extended states $(x_0, 0)$ and $(x_0, 1)$ have equal observations:

$$\pi_{\mathsf{M}^*}(x_0, 0) = \pi_{\mathsf{M}}(x_0) = \pi_{\mathsf{M}^*}(x_0, 1).$$

The extended observation map is noninjective. An exact latent decoder would imply injectivity. Therefore no exact latent decoder exists for M^*. □

I.1 Recovery from possible outputs

The extension and its base agree on the entire reachable set. Every function of the observation, including the identity, gives the same possible outcomes. The extension always lacks an exact decoder. The base may admit one.

Uniqueness of a latent representation requires a restriction on the admissible models or a further measurement. Dynamics, causal interventions, regularity, minimality, prior probabilities, and known state counts give different restrictions. Their justification depends on the application.

I.2 Physical interpretation

The Boolean label provides a second representation consistent with the same outputs. A physical interpretation of that label requires independent evidence. Approximate reconstruction would require a loss function, and probabilistic reconstruction would require a law on states or observations. A refined measurement would define a new equivalence relation.

J. Assumption checks and counterexamples

A composite map, a probability law, or a record over time can impose different conditions on recovery. The examples also distinguish total decoders from decoders on the reachable range.

J.1 Reachable collisions in a composite map

Let $f : \{0,1\} \to \{0,1,2\}$ satisfy $f(0) = 0$, $f(1) = 1$. Let $g(0) = 0$, $g(1) = g(2) = 1$. The map g is noninjective, but $g \circ f = \mathrm{id}_{\{0,1\}}$. Its collision lies outside the restriction of g to the reachable set $f[X]$. Conversely, if distinct original states satisfy $f(x) = f(y)$, then $g(f(x)) = g(f(y))$ for every downstream function g. This proves the corrected composition statement.

J.2 Decoder status of an already noninjective base

Take $X = \{0,1\}$, $O = \{0\}$, and $\pi(x) = 0$. No exact decoder exists. Its Boolean extension also has a constant observation map and no decoder. Thus the extension theorem guarantees failure of decoding for the extension, while a change of decoder status additionally requires a decodable base.

J.3 Empty spaces and total decoders

If $O = \varnothing$, existence of $\pi : X \to O$ forces $X = \varnothing$. The empty decoder is a left inverse. If $X = \varnothing$ but $O \neq \varnothing$, no total map $d : O \to X$ exists, although π is injective. If $X \neq \varnothing$ and π is injective, fix $x_0 \in X$. Define $d(o)$ as the unique preimage for reachable o, and as x_0 otherwise. Then $d \circ \pi = \mathrm{id}_X$. These cases justify the exact-decoder condition and the nonempty assumption in the global nonidentifiability theorems.

J.4 Equal possible outputs and different probability laws

For $0 < p < 1$, the Bernoulli law has reachable outcomes $\{0,1\}$. In particular, $p = 1/10$ and $p = 9/10$ share this set but differ in $\Pr(Y = 1)$. Any extension of empirical equivalence to statistical experiments must include the probability law. Equality of reachable sets alone supplies no statement about equal frequencies or likelihoods.

J.5 Temporal reconstruction and conditioning

For $\dot{x}_1 = x_2$, $\dot{x}_2 = 0$, and $y = x_1$, the exact solution is

$$y(t) = x_1(0) + t x_2(0).$$

At time zero, all values of $x_2(0)$ share one output for a fixed $x_1(0)$. The two-sample map is

$$\begin{pmatrix} y(0) \\ y(T) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & T \end{pmatrix} \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix}.$$

Its determinant is T, so it is invertible when $T \neq 0$, with

$$x_1(0) = y(0), \qquad x_2(0) = \frac{y(T) - y(0)}{T}.$$

If the two output errors are bounded by η, the recovered x_2 error is at most $2\eta/|T|$. Opposite signed endpoint errors attain the bound. Invertibility alone therefore gives no uniform stability as $T \to 0$. The regression test uses rational arithmetic, so it checks the exact formulas without integration or floating point tolerances.

K. Computational methods

The calculation enumerates the latent states, outputs, and pairs of states in each finite model.

K.1 Implementation and reproducibility

The Python program evaluates each processing rule on every reachable observation. It also computes the observation fibers and the refinement maps. The supplement includes source excerpts and the filenames of the complete records.

K.2 Finite models, fibers, and collision counts

The finite models use integer observation labels. We compare their reachable sets in the common space $O = \{0,1,2,3,4,5\}$, including the modulo-three outputs $0,1,2$ by their usual inclusion. The primary model has latent states

$$X = \{0,1,2,3,4,5\}$$

Table 2. Computational outputs used in the finite study.

Output group	Contents
Model summaries	Latent counts, reachable observations, fiber statistics, collision counts, injectivity, exact decoder status, and hidden multiplicity.
Experiment summaries	Possible outcome sets and equality with the injective base for four observation only experiments.
Predicate summaries	Existential success profiles and complete finite truth profiles for three accessible predicates.
Empirical quotient	Grouping of models by equality of reachable observation sets and comparison of latent properties within one class.
Refinement summaries	Interface sizes, strictness, collapsed pairs, reverse refinement, and induced quotient map properties.
Validation	Tests of the reference implementation, scope conditions, and measurement criteria.

and the identity observation map. A comparison model uses the same latent states and the rule $\pi(x) = x \bmod 3$. For every reachable observation o, the implementation enumerates the fiber

$$F_o = \pi^{-1}(o)$$

and calculates the unordered collision count

$$C(\mathsf{M}) = \sum_{o \in \mathscr{O}(\mathsf{M})} \binom{|F_o|}{2}.$$

This sum counts unordered pairs of distinct states with the same observation.

K.3 Hidden multiplicity family

The Boolean extension attaches $\{\text{false}, \text{true}\}$ to each base state and ignores the added coordinate during observation. The finite family attaches

$$H_m = \{0, 1, \ldots, m-1\}, \qquad m \in \{1, 2, 3, 5, 8\},$$

with observation rule

$$\pi_m(x, h) = x.$$

For the six state injective base, the finite identities are

$$|X_m| = 6m, \qquad |\mathscr{O}(\mathsf{M}_m)| = 6, \qquad |F_o| = m, \qquad C(\mathsf{M}_m) = 6\binom{m}{2} = 3m(m-1).$$

An exact decoder exists in this family only for $m = 1$.

K.4 Observation only experiments and predicates

Four rules process the observation: identity, a threshold at three, a partition into three categories, and a partition into bins of width two. Each rule acts on every reachable output.

The three predicates test evenness, $o \geq 3$, and membership in $\{0, 5\}$. The existential profile records whether each predicate succeeds for at least one output. The truth table records its value on every reachable output.

K.5 Finite empirical quotient and interface refinement

The finite empirical quotient groups models by exact equality of reachable observation sets. The grouping rule does not compare latent cardinality, fiber size, decoder status, or hidden factor metadata.

The refinement study uses one eight state world and four interfaces: identity, paired buckets, binary regions, and a relabeled identity. Each proposed refinement is checked on every world state. The implementation then records strictness, collapsed state pairs, reverse refinement, equality of observational resolution, and surjectivity, injectivity, and bijectivity of the induced finite quotient map.

K.6 Automated validation

The reference implementation has eighteen tests. Six further tests check assumptions about composition, decoding, probability, and temporal records. Two tests enumerate all reachable sets in codomains of size zero through three and all 4,096 triples of binary maps on four states. The suite also checks hidden extensions, predicate profiles, quotient grouping, reverse refinement, relabeling, and invalid coarsening. The program stops if any finite identity fails.

K.7 Formal proofs and finite calculations

Lean checks the proofs from the encoded definitions. Python enumerates the finite examples and checks each count against those definitions. The supplement identifies the Lean declarations and the source of the finite records.

Applying a proof to a physical device requires a model of its state, evolution, and readout (Horsman et al., 2014; Horsman, 2015). Measurement noise, approximation error, and omitted variables affect that application.

L. Computational results

The finite calculations satisfy the stated identities and invariance checks. The tables report exact enumerations, and the figures display the corresponding results for the multiplicity and refinement families.

L.1 Base models and Boolean extension

The injective base contains six latent states and six reachable observations. Every fiber has size one, the observation map is injective, and an exact decoder exists. The noninjective comparison contains six latent states and three reachable observations. Each reachable fiber has size two, the model contains three unordered latent collisions, and no exact decoder exists. The Boolean extension contains twelve latent states and the same six reachable observations as the injective base. Its maximum fiber size is two, it contains six collision pairs, and no exact decoder exists.

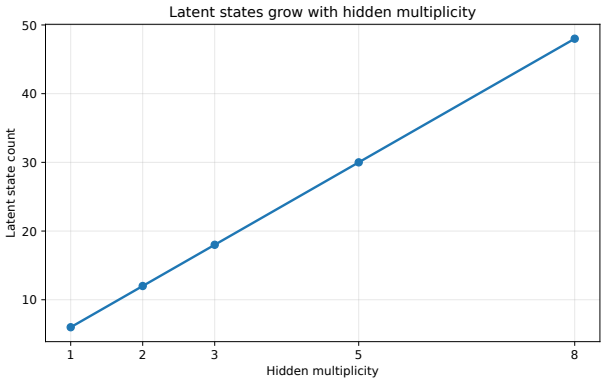
L.2 Finite hidden multiplicity

The latent state count increases linearly with hidden multiplicity. The reachable observation count remains fixed at six, and the maximum fiber size equals the hidden multiplicity. The four panels show the latent-state, reachable-observation, fiber-size, and experiment-outcome consequences of the same multiplicity construction.

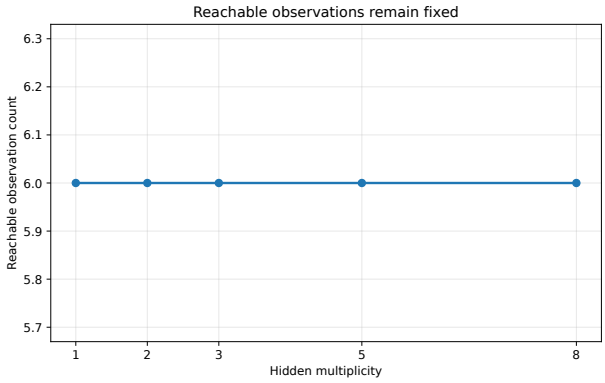
Every row is marked empirically equivalent to the base. Latent cardinality, fiber size, collision count, injectivity, and exact decoder status change while the reachable observation set remains fixed.

Table 3. Base and Boolean extension calculations.

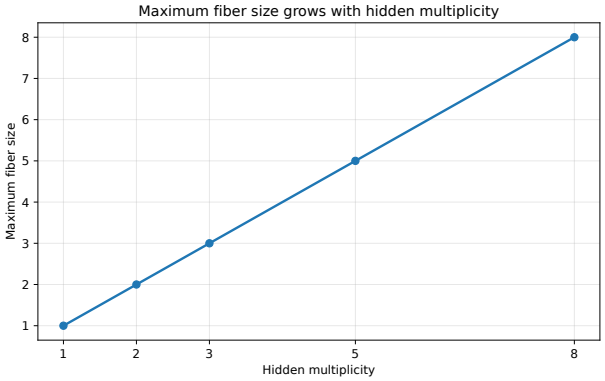
Model	Latent	Reachable	Max. fiber	Collisions	Injective	Exact decoder
Injective base	6	6	1	0	True	True
Noninjective base	6	3	2	3	False	False
Boolean extension	12	6	2	6	False	False



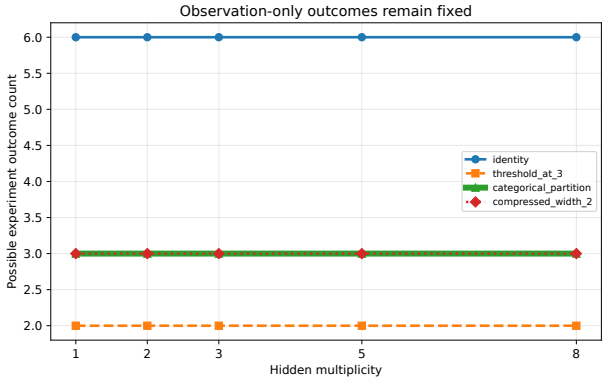
(a) Latent state count.



(b) Reachable observation count.



(c) Maximum fiber size.



(d) Observation only outcome counts.

Figure 6. Hidden multiplicity and observation only experiment results. The latent representation and observation fibers grow while the reachable observation set and each experiment's possible outcome count remain fixed.

Table 4. Complete finite multiplicity summary.

m	Latent	Reachable	Avg. fiber	Max. fiber	Collisions	Exact decoder
1	6	6	1.00	1	0	True
2	12	6	2.00	2	6	False
3	18	6	3.00	3	18	False
5	30	6	5.00	5	60	False
8	48	6	8.00	8	168	False

L.3 Observation only experiments and predicate profiles

The identity experiment has six possible outcomes. The threshold experiment has two. The categorical partition and width two compression experiments each have three. All twenty eight model and experiment comparisons match the base outcome set for the corresponding experiment.

Three predicates are evaluated for seven model representations. All twenty one existential comparisons match the base success profile. All twenty one complete truth profile comparisons also match the base.

Table 5. Observation only outcome sets.

Experiment	Possible outcomes	Count	All listed models match base
Identity	$\{0,1,2,3,4,5\}$	6	True
Threshold at three	$\{\text{false},\text{true}\}$	2	True
Categorical partition	$\{\text{high},\text{low},\text{middle}\}$	3	True
Compression width two	$\{0,1,2\}$	3	True

Table 6. Base predicate profiles and comparison counts.

Predicate	Base truth table on $0,\ldots,5$	Models compared	Matches
Even	T, F, T, F, T, F	7	7
At least three	F, F, F, T, T, T	7	7
Endpoint $\{0,5\}$	T, F, F, F, F, T	7	7

Table 7. Finite empirical quotient.

Class	Reachable observation set	Latent cardinalities	Representative types
1	$\{0,1,2,3,4,5\}$	6, 12, 18, 30, 48	Base, Boolean, and finite multiplicity extensions
2	$\{0,1,2\}$	6	Noninjective comparison model

L.4 Finite empirical quotient and varying latent properties

The finite grouping procedure produces two empirical classes. The first contains the injective base, Boolean extension, and finite multiplicity extensions. Its reachable set is $\{0,1,2,3,4,5\}$, while its representatives have latent cardinalities $6,12,18,30,48$. The second contains the noninjective comparison model with reachable set $\{0,1,2\}$.

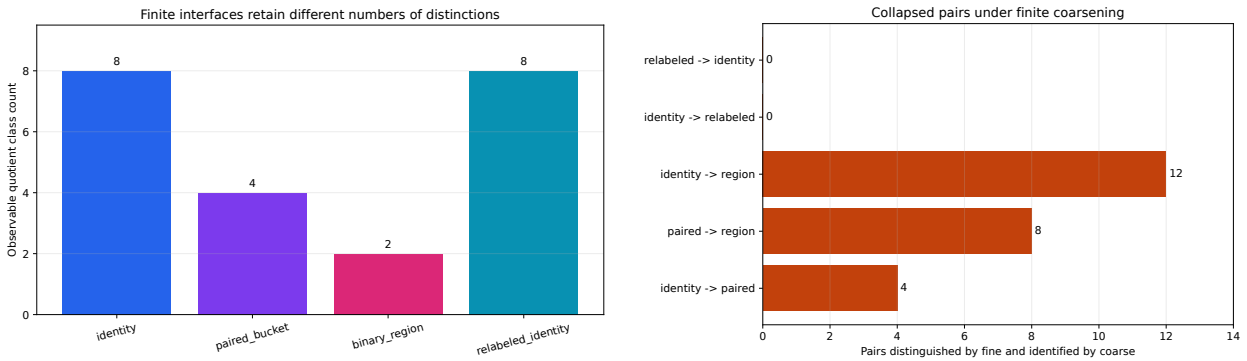
Four latent properties vary between the injective base and its Boolean extension although the two models occupy the same finite empirical class.

L.5 Interface refinement and quotient maps

The identity and relabeled identity interfaces retain eight observable classes. Paired buckets retain four. Binary regions retain two. The strict refinements collapse four, eight, and twelve world state pairs. Their quotient maps are surjective and noninjective, and no reverse refinement exists. The relabeling maps collapse no pairs and induce bijections.

Table 8. Properties that vary inside one finite empirical class.

Property	Base	Boolean extension	Varies
Latent cardinality	6	12	True
Maximum observation fiber size	1	2	True
Existence of exact latent decoder	True	False	True
Presence of Boolean hidden factor	False	True	True



(a) Observable quotient size.

(b) Collapsed state pairs.

Figure 7. Interface refinement results. Strict coarsening reduces the retained quotient classes and identifies world state pairs. Relabeling preserves observational resolution.

Table 9. Finite interface and refinement summary.

Refinement	Fine Q	Coarse Q	Collapsed	Strict	Surjective	Injective
Identity to paired	8	4	4	True	True	False
Paired to region	4	2	8	True	True	False
Identity to region	8	2	12	True	True	False
Identity to relabeled	8	8	0	False	True	True
Relabeled to identity	8	8	0	False	True	True

The files quotient_maps.csv and collapsed_pairs.csv contain the 36 entries in the quotient maps and the 24 collapsed pairs, respectively.

L.6 Validation result

All eighteen reference tests and eight additional finite checks pass. Each of the twelve regenerated CSV files matches its archived counterpart byte for byte.

M. Operational architecture

M.1 Process, feedback, and perspectival description

An observation can form one step in a feedback process. Whitehead treated events and relations as basic units of description (Whitehead, 1978). Wiener studied how communication and control depend on feedback (Wiener, 1948). Von Neumann analyzed how measurement and self-reproduction couple a system to an apparatus or to a description of itself (von Neumann, 1955, 1966). Leibniz considered how one world can admit different perspectives (Leibniz, 1989).

Bohm distinguished implicate from explicate order. Bohm and Hiley developed an account of quantum process on that basis (Bohm, 1980; Bohm and Hiley, 1993). The observer model requires only a map between sets.

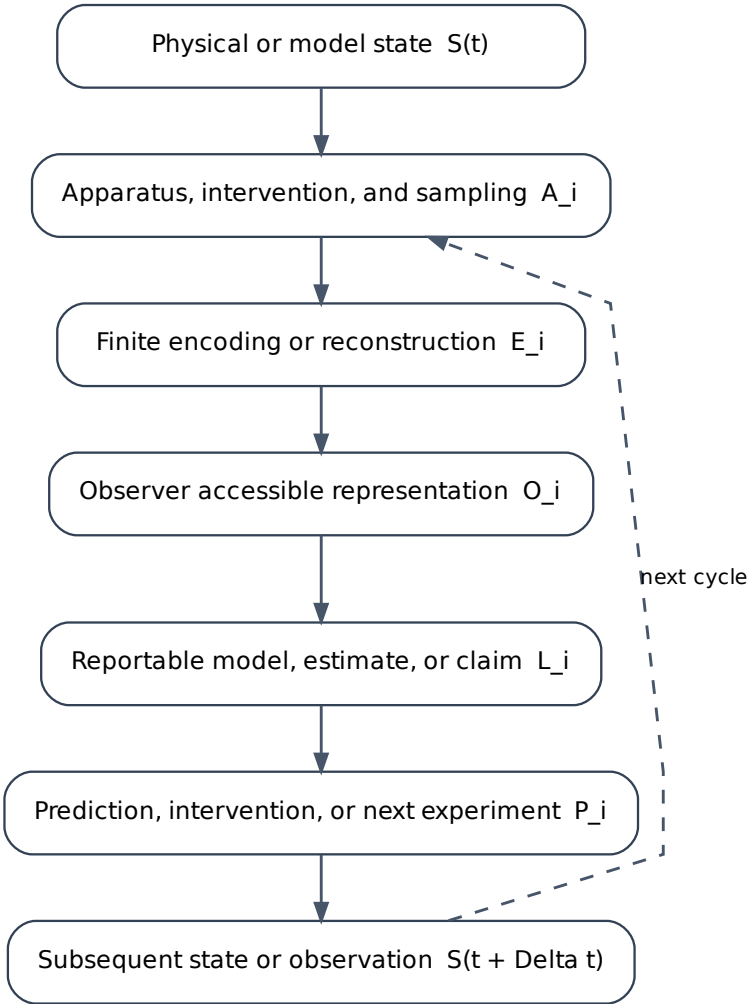


Figure 8. Observation and feedback. The dashed return connects a prediction or intervention to the next observation.

M.2 The complete recursive scientific cycle

A measurement may guide the next action on a system. Write $S(t)$ for the system state, A_i for the apparatus or intervention, and E_i for an encoding. The observer has access to O_i and can produce a report, model, or estimate L_i. The resulting prediction or action P_i can affect the next measurement.

$$S(t) \longrightarrow A_i \longrightarrow E_i \longrightarrow O_i \longrightarrow L_i \longrightarrow P_i \longrightarrow S(t+\Delta t).$$

After the action, the system enters the next measurement cycle. Predictive processing and active inference likewise relate the update of a model to actions that affect later observations (Rao and Ballard, 1999; Friston, 2010; Parr et al., 2022; Friston et al., 2025).

Each arrow requires a map. For example, an instrument converts a physical state into a reading, and a reconstruction procedure converts that reading into a feature.

M.3 A distributed family of boundaries

A sequence of observation and action maps has the form

$$X_{\text{phys}} \xrightarrow{\pi_{\text{app}}} O_{\text{app}} \xrightarrow{\pi_{\text{bio}}} O_{\text{bio}} \xrightarrow{\pi_{\text{rep}}} O_{\text{rep}} \xrightarrow{e} R \xrightarrow{a} X'_{\text{phys}}.$$

The maps may correspond to physical coupling, detector response, sampling, reconstruction, biological sensing, attention, memory, language, or experimental decision. The aggregate observation map is

$$\pi = \pi_{\text{rep}} \circ \pi_{\text{bio}} \circ \pi_{\text{app}}.$$

If two distinct original source states reach the same intermediate value, every subsequent deterministic map preserves that equality. The composite is then noninjective. Noninjectivity of a stage outside the range reached by preceding stages is insufficient; a counterexample is given in Appendix J. Refining a later report cannot recover a distinction that was already collapsed upstream unless new information enters through another route.

Modeling each stage separately identifies where two source states first give the same output. A later measurement can distinguish the states only if another channel carries information about their difference.

M.4 Computation as an interface-bound physical process

A physical computer requires an encoding, an evolution rule, and a readout:

$$S_{\text{phys}} \xrightarrow{\eta} C \xrightarrow{U} C' \xrightarrow{\rho} R.$$

Here η assigns an abstract computational state to a physical condition, U is the declared abstract evolution, and ρ converts a final physical or encoded state into a reportable result. Frameworks for physical computation make this representation relation explicit and distinguish the abstract calculation from the device that realizes it (Horsman et al., 2014; Horsman, 2015). When the computation is embedded inside the system being studied, both η and ρ are themselves observer interfaces.

Physical computers differ in their state spaces, available operations, error models, and readouts. A digital computer represents finite symbolic states through engineered voltage ranges. An analog computer uses calibrated continuous physical variables to approximate a differential relation (Shannon, 1941). A quantum computer manipulates quantum states but ultimately requires a controlled physical implementation and a classical measurement interface (DiVincenzo, 2000; Aharonov et al., 2022). A brain–computer interface converts selected electrophysiological activity into device commands and requires adaptation of both the user and decoder (Wolpaw et al., 2002; Willett et al., 2023). Human reasoning is mediated by sensory transduction, neural dynamics, memory, language, and report. Equality of an output across these systems does not imply equality of their internal logic, physical representation, or relation to the world.

Table 10. Physical interfaces, their accessible variables, and sources of uncertainty.

Interface	Accessible state and readout	Principal boundary
Digital computer	Finite symbolic words encoded by engineered voltage ranges and returned through a discrete output channel.	Word length, architecture, numerical precision, software semantics, and the mapping between physical voltage and symbol.
Analog computer	Continuous voltages, currents, rotations, or other calibrated quantities interpreted as variables in a differential model.	Noise, drift, dynamic range, calibration, and the approximation relating physical trajectory to mathematical solution.
Quantum computer	Prepared quantum states, controlled transformations, and measurement outcomes converted into classical records.	State preparation, decoherence, permitted measurements, sampling, error correction, and the quantum–classical readout boundary.
Brain–computer interface	Selected neural signals translated into commands, text, movement, or another device output.	Electrode placement, signal preprocessing, decoder training, adaptation, and the distinction between intended state and decodable signal.
Human sensory and cognitive interface	Receptor signals integrated through neural dynamics, memory, attention, language, and report.	Biological bandwidth, selective attention, prior structure, state dependence, and reportability.
Multisensor suite and fused state estimate	Heterogeneous measurements combined with a dynamical model or reference data into an estimated state, residual, confidence value, or track.	Coverage, calibration, synchronization, bias, correlated error, fusion rule, model mismatch, and unobserved state components.
Inertial guidance and state estimation	Accelerometer and gyroscope signals combined with a dynamical model to maintain an estimated position, velocity, attitude, and error state.	Sensor bias, drift, model mismatch, initialization, reference updates, and uncertainty in the estimated state.
Particle detector and analysis pipeline	Detector interactions reconstructed into tracks, energies, event classes, likelihoods, and parameter limits.	Acceptance, efficiency, resolution, trigger selection, reconstruction assumptions, background model, and statistical decision rule.

Each sensor measures part of the state of a vehicle or its environment. Combining readings can improve an estimate. Its uncertainty depends on coverage, calibration, timing, and the fusion rule (Hall and Llinas, 1997; Durrant-Whyte and Henderson, 2008).

An inertial guidance system compares predicted measurements with sensor readings. The residual is the difference between the prediction and the reading. Interpreting it as a state error requires a dynamical model and estimates of measurement uncertainty (Kalman, 1963; Hall and Llinas, 1997; Titterton and Weston, 2004).

In particle physics, a theory can forbid a process under its kinematic assumptions. A detector can also miss a permitted process because of its acceptance or a blind region. A statistical exclusion

Table 11. Taxonomy of outcomes across an observation or representation map.

Outcome	Structural meaning	Mathematical or empirical question
Preservation	A relation, value, or invariant survives the map.	Which quantities factor through the interface, and are they complete for the target task?
Re-encoding	Structure is expressed in a different representation.	Is the transformation invertible, approximately invertible, or only relation-preserving?
Compression	The accessible representation contains fewer distinguishable classes than the source.	Which fibers are enlarged, and what refinement would split them?
Dispersion	Correlations or organized structure spread across degrees of freedom, time, or scale.	Which observables still carry the relation, and at what scale or resolution?
Collapse	Distinct source states map to one accessible state.	Is the map noninjective, and what latent properties vary inside a fiber?
Inaccessibility	A structure may remain in the source but lies outside the declared observation channel or threshold.	Does another interface represent it, or is its existence itself underdetermined?
Physical destruction	The source structure no longer exists under a domain-specific criterion.	What physical law, measurement, or dynamical model establishes destruction rather than nonrepresentation?

depends on the data, background model, reconstruction, and confidence procedure (Cowan et al., 2011; Particle Data Group, 2026).

A proof checker establishes what follows from encoded premises under its inference rules. Relating those premises to a physical sensor requires a model of the sensor and its readout.

M.5 Preservation, transformation, compression, dispersion, and collapse

A measurement can preserve a target quantity while losing distinctions in other variables. Table 11 distinguishes several possible outcomes.

A quantity can survive a change of representation. A correlation can spread across channels until one reading no longer detects it. Physical destruction requires a criterion for disappearance of the structure in the source.

M.6 External analyst and embedded observer

An external analyst specifies the source space and the map to observations. An embedded observer forms its self-model inside the process it observes.

An embedded model must account for how an observation changes the observer that made it. Second-order cybernetics, autopoiesis, active inference, and work on self-referential living systems address such feedback (von Foerster, 1981; Maturana and Varela, 1980; Friston, 2010; Abramsky et al., 2025).

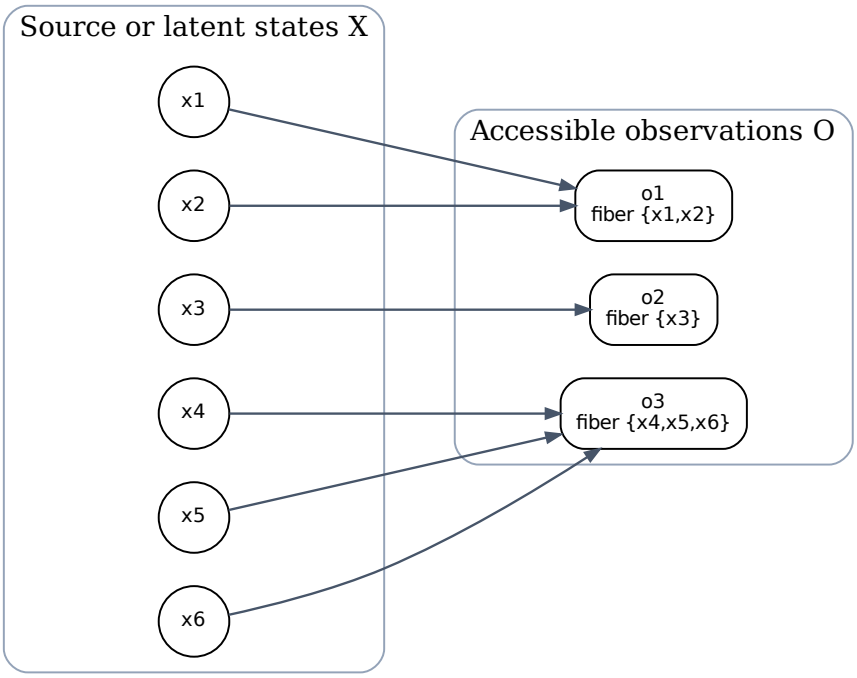


Figure 9. Observation fibers. States in one fiber produce the same accessible value. Fiber membership is relative to the specified map. A refined interface may split a fiber that a coarse interface leaves merged.

M.7 Reduction to the formal core

A single observation and its processing have the form

$$X \xrightarrow{\pi} O \xrightarrow{e} R,$$

where X is the source space, O the observation space, and $e : O \to R$ the rule that processes an observation. The map π may represent one measurement or a composition of several maps.

M.8 Three levels of description

A source state belongs to X, an observation to O, and an experimental result to R. Several states can give one observation, and several observations can give one result. Two models are empirically equivalent when they have the same set of reachable observations.

M.9 Local noninvertibility

For $o \in O$, the fiber is

$$\pi^{-1}(o) = \{x \in X : \pi(x) = o\}.$$

When this set contains multiple states, the observation identifies the fiber but not a unique member.

Exact recovery requires an injective map on the admissible state class. An additional measurement or a justified physical restriction can separate states inside an original fiber.

N. Related work

Which properties of a system can an observer recover from a measurement? Observability, statistical inference, and theories of perception approach this question with different definitions of a measurement. Each theory specifies what the observer can record.

N.1 Observability and identifiability

In control theory, an output trajectory can carry information about a state that a single reading omits. Kalman studied state recovery in linear systems (Kalman, 1963). Hermann and Krener studied observability in nonlinear systems (Hermann and Krener, 1977). In structural identifiability, the unknown quantity may be a parameter or a model structure (Bellman and Åström, 1970; Rothenberg, 1971; Walter and Pronzato, 1997). Two parameter choices can agree on every ideal observation. Noise and limited sampling can also make distinguishable parameters difficult to estimate (Calderón, 1980; Raue et al., 2009; Chis et al., 2011; Villaverde et al., 2016; Meshkat et al., 2009; Guillaume et al., 2019).

A nontrivial observation fiber gives a specific form of nonuniqueness. Several states produce the same reading. Well-posedness also requires existence and stability of the solution (Hadamard, 1923; Tikhonov, 1963; Engl et al., 1996). Regularization can select one state in a fiber. A claim that the measurement identifies that state requires an additional assumption or observation.

Latent-variable models likewise require conditions for uniqueness (Teicher, 1963; Kruskal, 1977; Allman et al., 2009; Anandkumar et al., 2014). In unsupervised learning, distinct latent representations can agree on the observations (Locatello et al., 2019). Auxiliary variables, multiple views, and restrictions on the model can restore identifiability in specified settings (Hyvärinen et al., 2019; Khemakhem et al., 2020; Gresele et al., 2020; Xi and Bloem-Reddy, 2023). Tenenbaum and colleagues describe how learners acquire structured abstractions from data (Tenenbaum et al., 2011). In the deterministic model, adding an unread coordinate gives an explicit example of two latent representations with the same reachable observations.

N.2 Equivalence under a family of tests

The information available to an experiment determines which systems it can distinguish. Blackwell compares statistical experiments by their informativeness for decisions (Blackwell, 1951, 1953; Torgersen, 1991). Moore experiments and bisimulation compare states by specified external behavior (Moore, 1956; Park, 1981; Milner, 1989; Larsen and Skou, 1991; Rutten, 2000). In causal analysis, interventions can separate graphs that have the same observational conditional independences (Verma and Pearl, 1990; Chickering, 1995; Andersson et al., 1997; Pearl, 2009; Hauser and Bühlmann, 2012). An observation-only experiment in the deterministic model receives a single reachable output. Equality of reachable sets constrains those experiments. Comparing event probabilities or temporal records requires a model that assigns them.

N.3 Appearance, perspective, and hidden order

Philosophical accounts of perception distinguish an appearance from the conditions that produce it. Plato's cave illustrates how a display can reveal only part of its source (Plato, 1992). Aristotle distinguishes several senses of cause (Aristotle, 1984). Husserl asks how scientific objects acquire their meaning through intentional activity and the lifeworld (Husserl, 1970). In an observation

model, the distinction has a measurable consequence. Different source states can produce one accessible representation.

Leibniz asks what grounds a distinction between things and whether distinct entities can share all their properties (Leibniz, 1989; Leibniz and Clarke, 2000). His account of perspective gives each monad a particular view of one world. Restricting the question to properties accessible through a map changes its answer. Two states in one fiber agree on every predicate that depends only on the reading. A predicate defined on the source can still distinguish them.

Whitehead describes reality through processes and relations (Whitehead, 1978). Bohm's implicate order and his work with Hiley relate observable phenomena to a deeper physical process (Bohm, 1980; Bohm and Hiley, 1993). A hidden extension requires only a state coordinate omitted by the readout. A physical interpretation of that coordinate requires a model of how the coordinate affects measurable behavior.

N.4 Formal limits and mechanical proof

Diagonal arguments place limits on what particular formal systems can express. Cantor constructs an object outside a proposed enumeration (Cantor, 1891). Russell identifies a contradiction in unrestricted set formation (Russell, 1908). Gödel establishes incompleteness under conditions on a formal theory (Gödel, 1931). Tarski gives conditions under which truth cannot be defined within the same language (Tarski, 1936). Lawvere formulates a categorical fixed-point principle (Lawvere, 1969). Hofstadter explores self-description in formal systems and minds (Hofstadter, 1979, 2007). Penrose proposes a disputed connection between Gödelian limits and human cognition (Penrose, 1989, 1994). For a measurement, the obstruction to exact state recovery can be tested by asking whether two allowed states give the same reading.

Leibniz proposed a symbolic calculus for reasoning (Leibniz, 1989). Modern proof assistants check whether a conclusion follows from encoded premises and inference rules. Lean and mathlib support this kind of verification (de Moura et al., 2015; The mathlib Community, 2020). The Flyspeck project applied mechanical checking to a geometric proof (Hales et al., 2017). A proof about a readout establishes a property of the specified map. Applying it to an instrument also requires evidence that the map describes the instrument's readings.

N.5 Feedback and embedded observation

An embedded observer can act on the system it observes. Wiener studied communication, control, and feedback (Wiener, 1948). Ashby related the variety of disturbances to the responses available to a regulator (Ashby, 1956). Second-order cybernetics includes the observer in the system under study (von Foerster, 1981). Autopoiesis and relational biology examine how a living system maintains its organization (Maturana and Varela, 1980; Rosen, 1991). Von Neumann's work on measurement and self-reproducing automata links physical operations to descriptions of those operations (von Neumann, 1955, 1966).

An external analyst can specify a readout without specifying its effect on the system. A model of an embedded observer must also describe how a reading changes later states or actions. Elshatlawy and colleagues propose operational equivalences and complexity measures for observers (Elshatlawy et al., 2025). Abramsky and colleagues identify time, self-reference, and self-modification as open problems for models of living systems (Abramsky et al., 2025).

N.6 Information, scale, and physical computation

Information theory quantifies distinctions that survive transmission and processing (Shannon, 1948a,b; Kullback and Leibler, 1951; Cover and Thomas, 2006). Theories of self-organization and emergence address how constraints and stable structures persist across scales (Nicolis and Prigogine, 1977; Kauffman, 1993; Deacon, 2011; Mitchell, 2009). Hierarchical active inference constructs coarse states and paths across levels (Friston et al., 2025). Applying the deterministic quotient at two scales requires an explicit map between them. A quantity on the finer states descends to the coarser states exactly when it is constant on each fiber.

A mathematical invariant and its physical interpretation have separate conditions. Hilbert's axioms admit geometric realizations that satisfy the same relations (Hilbert, 1902). Newtonian mechanics and special relativity specify the changes of description under which their predictions agree (Newton, 1999; Einstein, 1905). Physical predictions can then be tested against measurements (Feynman, 1965). Wheeler considered the role of observation in physical description (Wheeler, 1983). Constructor theory expresses physical laws in terms of possible and impossible transformations (Deutsch, 2013; Deutsch and Marletto, 2015). Tegmark proposes information-processing properties for consciousness as a state of matter (Tegmark, 2015). Wolfram describes bounded observers that combine computational histories into a single description (Wolfram, 2002, 2023; Rickles et al., 2023). Each proposal assigns its own physical or computational meaning to an observation.

A device's encoding and readout determine which abstract calculation its physical evolution implements. Shannon's differential analyzer uses continuous physical evolution for analog calculation (Shannon, 1941). Horsman and colleagues make the relation between physical states and abstract computation explicit (Horsman et al., 2014; Horsman, 2015). Quantum computation distinguishes preparation, controlled evolution, measurement, and classical output (DiVincenzo, 2000; Aharonov et al., 2022). Brain–computer interfaces use selected neural signals to generate device commands (Wolpaw et al., 2002; Willett et al., 2023). Two devices can thus agree on the output of a computation while differing in substrate, noise, or accessible internal state.

N.7 Consciousness, agency, and report

A report is one observation of a subject under specified conditions. Chalmers distinguishes functional accounts from an explanation of subjective experience (Chalmers, 1995). Theories of integration, global availability, attention, resonance, and predictive processing propose different relations among neural activity, report, and experience (Tononi, 2008; Koch, 2019; Graziano, 2013; Grossberg, 2021; Friston, 2010; Parr et al., 2022). Jaynes proposed a historical account of self-modeling and source attribution (Jaynes, 1976). Hoffman and Prakash distinguish perceptual interfaces from the objective structures that produce their outputs (Hoffman and Prakash, 2014; Prentner and Hoffman, 2024). Theories of agency also distinguish levels of description. Sapolsky emphasizes the causal history of an action. List, Dennett, and Frankfurt describe forms of agency compatible with physical causation (Sapolsky, 2023; List, 2019; Dennett, 2003; Frankfurt, 1971).

An experiment distinguishes theories when they predict different accessible outcomes (Gomez-Marin and Seth, 2025; Tononi et al., 2025). The COGITATE collaboration preregistered divergent predictions of integrated information theory and global neuronal workspace theory. Its results challenged predictions of both (Cogitate Consortium et al., 2025). Laukkonen, Friston, and Chandaria propose a recursively shared world model and precision control as conditions for consciousness

(Laukkonen et al., 2025). Whyte and colleagues formulate a minimal testable theory within active inference (Whyte et al., 2026). Robinson and colleagues propose comparing active gaze shifts with passive replay of the same visual displacement (Robinson et al., 2025). The assigned condition supplies a record separate from the participant's report that the target reappeared.

Artificial systems can expose different information through a verbal report, an internal state, and a monitoring channel. Chalmers examines possible routes to consciousness in language models (Chalmers, 2023). Butlin and colleagues propose indicators drawn from neuroscientific theories (Butlin et al., 2023). Testing an indicator requires a channel that distinguishes the internal states on which the indicator takes different values.

N.8 Neural manifolds and biological measurements

Grid-cell studies recover population geometry from recordings and behavioral measurements (Gardner et al., 2022). Related studies examine toroidal organization during one-dimensional behavior and the role of oscillatory dynamics in the recovered topology (Hermansen et al., 2024; di Sarra et al., 2025a). Spike sorting, behavioral sampling, and dimensionality reduction each restrict the available record. A published correction replaced duplicated panels in the synthetic torus calibration figure of the oscillation study (di Sarra et al., 2025b). The robustness of a recovered neural manifold can be tested by varying the sampling and preprocessing and comparing the resulting topology.

Ultraweak photon emission supplies a different biological measurement. Research reports spontaneous emission from DNA under controlled conditions, differences between living and dead mice, and stress-related changes in plants (Pietruszka and Marzec, 2024; Salari et al., 2025). Detection methods and biological applications depend on the sources and intensities of the emissions (Mould et al., 2024). The hypothesis that light carries information about living organization requires a specified property of that organization and a test of which emission readings distinguish its values.

N.9 Observable state and empirical model class

An observation map partitions states into fibers. A target can be recovered from a reading exactly when it is constant on the corresponding fiber. At the model level, equality of reachable sets groups models that admit the same observation labels. An unread coordinate can change the latent structure and prevent exact state recovery while preserving this reachable set. A second measurement improves recovery precisely when it separates the states that disagree on the target.

O. Broader applications and research hypotheses

The observation model compares exact outputs of a fixed deterministic map. Applications to living systems require measurements of time, noise, feedback, or physical organization. Each application also needs an observable that distinguishes its hypothesis from an alternative.

O.1 Set-based empirical equivalence and exact equality

Empirical equivalence is equality of reachable observation sets. Real measurements include probabilities, frequencies, noise, finite precision, and time dependence. Two models can have the same support and different probability laws. A stochastic extension should assign measures or kernels to latent and observation spaces. Approximate equivalence should replace exact fibers by metric neighborhoods, couplings, or reconstruction-error bounds.

O.2 Static interfaces and unrestricted hidden factors

Two states can give the same instantaneous reading and different trajectories. A dynamical model must specify evolution, inputs, interventions, and the recorded outputs. Physical laws, regularity, causal structure, or a minimality condition may rule out hidden extensions that the static model allows.

O.3 Composite chains and local loss analysis

Apparatus, sampling, encoding, perception, and reporting define stages of an observation. Their composite describes what reaches the final report. Examining each stage can locate the first loss of a source distinction. Adding a new measurement can restore a distinction when that measurement reaches the source by another route.

O.4 The embedded self-interface

An external analyst supplies the source space and observation map. An embedded system forms a self-model and uses its own observations to update its state. Work on feedback, self-production, self-reproducing automata, and computational observers examines parts of this problem (Wiener, 1948; von Foerster, 1981; Maturana and Varela, 1980; Rosen, 1991; von Neumann, 1966; Wolfram, 2023; Rickles et al., 2023). One such update has the form

$$S_t \xrightarrow{\pi_{\text{self}}} O_t^{\text{self}} \xrightarrow{u} S_{t+1}.$$

Can the system justify its observation map and update rule using its own resources? Coalgebra, domain theory, second-order cybernetics, and active inference offer candidate models for that question (von Foerster, 1981; Abramsky et al., 2025; Friston et al., 2025).

O.5 Cross-domain candidate invariant

The conjecture about an observer that studies its own means of observation is

> *No sufficiently expressive representational system can completely justify or perceive the boundary that generates its own representational capacity.*

Testing the conjecture requires a definition of the system, its boundary, its representations, and what would constitute complete justification. Results on formal logic, observer models, and living systems provide separate test cases (Gödel, 1931; Tarski, 1936; Lawvere, 1969; Elshatlawy et al., 2025; Abramsky et al., 2025).

O.6 Paradox as a boundary signal

Hypothesis O.1 (Boundary-signal hypothesis)**.** Some recurrent paradoxes may arise when a representational system encounters a limit on what it can express about itself.

A test would distinguish inconsistent premises, ambiguous terms, undecidable statements, underdetermined observations, and changes in access. A defined measure of the boundary would have to predict where such failures occur. Cases explained entirely by inconsistent premises or error count against the hypothesis.

O.7 Observer-relative mathematics

A further philosophical hypothesis is that mathematics arises through interaction between an observer and reality, while shared mathematics records structures that remain invariant across

different representational routes. In its strongest informal form, "what we call mathematics is what observers can preserve and agree upon." This is not implied by the formal theorem. Mathematical structuralism and category-theoretic invariance provide relevant comparison points (Shapiro, 1997; Leinster, 2014). Hilbertian axiomatics separates a structure from one concrete realization, while Newtonian and Einsteinian physics illustrate how stable law or invariant relation can survive changes of coordinates and representation (Hilbert, 1902; Newton, 1999; Einstein, 1905). Feynman's account of physical law supplies a further methodological reminder that mathematical form earns physical force only through contact with experiment (Feynman, 1965). A future paper would need to distinguish conventional notation, structural equivalence, proof-theoretic invariance, and observer-dependent access to mathematical objects.

O.8 Materialism, constraint, and unified-state formation

Hypothesis O.2 (Constraint and unified-state hypothesis)**.** A physical account of consciousness must explain which constraints allow high-dimensional activity to sustain a stable, integrated state that a subject can report.

The proposed target is the constraint that lets a system behave as one state over time. Hard determinist accounts emphasize the causal history of biological action. Compatibilist accounts examine reasons, alternatives, control, and endorsement (Sapolsky, 2023; List, 2019; Dennett, 2003; Frankfurt, 1971). Studies comparing active inference, global workspace theory, and integrated information theory offer ways of testing distinct mechanisms (Seth and Bayne, 2022; Cogitate Consortium et al., 2025; Laukkonen et al., 2025; Whyte et al., 2026; Robinson et al., 2025). Such a comparison requires a measurement on which the proposed mechanisms differ.

A constraint measure would support the hypothesis if it improved predictions of transitions in reportability during sleep, anesthesia, perturbation, or pathology. A model of the component processes that predicts the same transitions without the added variable would count against it.

O.9 Scale-conditioned death and observability

Hypothesis O.3 (Scale-transition hypothesis of death)**.** After death, some correlations present at the organismal scale may persist at other spatial or temporal scales even when organismal organization is no longer detectable.

The physical targets differ. Matter and energy can persist after death. The persistence of a specified correlation, the persistence of organismal organization, and the continuity of personal identity each require a separate criterion.

A test would define a measure of coherence at the organismal scale and record selected correlations before and after death. Repeating the measurements at several resolutions would reveal which correlations remain detectable. A claim that they preserve organismal organization requires a reconstruction criterion.

O.10 Light as an informational standard hypothesis

Hypothesis O.4 (Light and living organization)**.** "Light is the informational standard of the living; when it turns into something else it is the informational standard of the dead."

A measurable version of this hypothesis needs an emission statistic and a specified change in that statistic. Candidate changes include a spectral shift, a change in temporal correlation, spatial dispersion, or coupling to another physical variable.

Ultraweak photon emission is one candidate measurement. Studies report differences in living and dead mice and measurements in DNA, cells, and stressed plants (Pietruszka and Marzec, 2024; Salari et al., 2025; Mould et al., 2024). A test would ask whether a specified emission statistic predicts organismal organization or viability after accounting for temperature, oxidative state, metabolism, and tissue composition.

O.11 Single-observer, intentional-forgetting, and external-source hypotheses

One metaphysical hypothesis proposes that a foundational observer generates a contained reality and relinquishes access to its own control or memory. On this account, forgetting permits uncertainty, discovery, and choice in encounters with apparently independent beings. A related hypothesis proposes that an external structure leaves observable traces through boundary conditions, invariants, or failures of closure.

Evidence for such a hypothesis would require a measurable difference from the alternatives.

A test must name an outcome, give its predicted probability under each model, or identify an output permitted by only one model. If physical, plural-observer, information-based, and cognitive accounts give the same predictions, the observed report leaves the external-source hypothesis undecided.

O.12 Containment, internal reality, and meaning

A philosophical hypothesis about containment concerns which relations persist across observers. Experiences and relations could be meaningful within a containing process even if their existence depends on it. Leibniz, Whitehead, and Peirce offer different accounts of perspective, relation, and fallible inquiry (Leibniz, 1989; Whitehead, 1978; Peirce, 1877, 1878). A mathematical formulation would specify a family of transformations and test which relations they preserve.

O.13 Identity, factorial zero, layered tensors, and geometry

The unit value 1 occurs in 0!, the empty product, e^0, $\Gamma(1)$, the determinant of an identity matrix, and probability normalization. A proposed analogy connects the unit with geometric structure:

$$\text{preserved unit} \longrightarrow \text{metric structure} \longrightarrow \text{connection} \longrightarrow \text{curvature}.$$

A geometric calculation requires a specified smooth metric. Its Levi-Civita connection depends on the metric and its first derivatives. Curvature depends on the connection and its derivatives (Nakahara, 2003). Even in flat space, a change of coordinates can produce nonzero connection coefficients. The value $0! = 1$ alone specifies neither a metric nor its curvature.

O.14 Neural manifolds and reportability

Grid-cell population studies recover geometry from a sequence of measurements and transformations (Gardner et al., 2022; Hermansen et al., 2024; di Sarra et al., 2025a). These include electrode signals, spike sorting, behavioral alignment, dimensional reconstruction, and topological summaries. A test of the recovered geometry can compare results across specified changes in sampling, preprocessing, or behavioral context. A claim about experience requires a further link to perception or report.

O.15 Heterogeneous computational and engineered interfaces

Digital, analog, quantum, neural, and engineered systems have different state spaces, operations, error models, and readouts. A comparison can represent each implementation by

$$(S_i, \eta_i, U_i, \rho_i)$$

and specify which encodings and readouts realize the same abstract function. The same comparison can determine whether one interface refines another.

Multisensor fusion combines partial readings into a state estimate whose uncertainty depends on coverage, calibration, timing, and the fusion rule (Hall and Llinas, 1997; Durrant-Whyte and Henderson, 2008). Inertial guidance uses estimates of position, velocity, and attitude (Kalman, 1963; Titterton and Weston, 2004). A particle-physics exclusion contour depends on detector acceptance, event reconstruction, a model, and a statistical rule (Cowan et al., 2011; Particle Data Group, 2026).

O.16 Open mathematical problems

Possible extensions of the observation map include the following.

(i) stochastic observation channels and equality of induced measures;
(ii) approximate equivalence and quantitative reconstruction error;
(iii) dynamical observability with inputs and interventions;
(iv) composition of multiple interfaces and stage-specific fiber analysis;
(v) information measures for quotient size, dispersion, and refinement cost;
(vi) criteria for minimal or canonical representatives;
(vii) embedded self-interfaces with feedback and model revision;
(viii) automated countermodel and discriminating-experiment construction;
(ix) formal comparison with Leibnizian indiscernibility and perspectival representation;
(x) computational-irreducibility and bounded-observer criteria;
(xi) artificial-system interfaces and machine-consciousness indicator tests;
(xii) translation maps among digital, analog, quantum, neural, and human computational interfaces;
(xiii) state-estimation systems whose actions depend on an internal model and quantified error;
(xiv) multisensor suites that combine partial observations into a bounded state estimate or decision;
(xv) particle-physics acceptance, blind-region, and exclusion analyses as distinct kinds of boundary claim; and
(xvi) domain-specific applications with independently justified maps.

O.17 Hypothesis and application register

Table 12. Status and next requirement for each broader application.

Program	Current status	Required next object	What would count as progress
Embedded boundary invariant	Cross-domain conjecture	Domain-specific definitions and a common translation theorem	A theorem recurring across at least two nontrivially different domains, or a counterexample that narrows the claim

Program	Current status	Required next object	What would count as progress
Paradox as boundary signal	Hypothesis	Classifier separating error, undecidability, and empirical underdetermination	Prospective prediction of boundary failures not explained by ordinary inconsistency
Observer-relative mathematics	Philosophical hypothesis	Invariance class across formal languages or observers	A precise theorem connecting translation-invariant structure to interface equivalence
Constraint and unified consciousness	Empirical and computational program	Defined constraint or integration variable and competing models	Preregistered predictions across state transitions that separate mechanisms
Scale-conditioned death	Empirical hypothesis	Coherence criterion across scales and time	Measured transformed structure with a justified continuity or reconstruction relation
Light as informational standard	Not yet operationalized	Spectral, temporal, spatial, or information statistic	Predictive value beyond metabolic, thermal, and oxidative confounds
Single foundational observer	Metaphysical hypothesis	Unique discriminating prediction	An outcome not shared by plural-observer, physical, information-only, or cognitive alternatives
External source or revelation	Metaphysical hypothesis	Publicly checkable boundary condition or invariant	Reproducible evidence with lower likelihood under internal alternatives
Identity-to-geometry motif	Separate mathematical note	Independent differential-geometric derivation	Correct theorem showing the precise relation, or a clear statement of the analogy's limits
Leibnizian perspective and indiscernibility	Historical-formal comparison	Translation between all properties and interface-accessible predicates	A theorem specifying when interface equivalence preserves or fails Leibnizian indiscernibility
Process and constraint dynamics	Dynamical application program	State evolution, constraints, and measurable order parameters	A model separating process, constraint, and observation effects with falsifiable predictions
Computational irreducibility and observer theory	Comparative formal program	Explicit coarse-graining and resource bounds	A theorem relating computational aggregation to interface refinement or a counterexample separating them

Program	Current status	Required next object	What would count as progress
Artificial-system observer interfaces	Empirical and ethical application	Declared outputs, hidden states, self-monitoring channels, and indicator criteria	Tests that distinguish report equivalence from mechanism or consciousness claims
Neural manifold and reportability	Empirical application	Explicit maps from neural activity through measurement and reconstruction to report	Results stable across justified preprocessing, sampling, reconstruction, and behavioral changes
Heterogeneous physical computation	Formal and engineering application	Encoding, evolution, and readout maps for each implementation	A theorem stating when digital, analog, quantum, neural, or human interfaces realize the same abstract relation
Inertial navigation and guidance	State-estimation application	Dynamical model, sensor map, error state, and uncertainty relation	Correct predictions of when distinct physical trajectories remain indistinguishable to the estimator
Multisensor suite and data fusion	Safety-critical state-estimation application	Sensor-specific observation maps, timing, uncertainty, and a declared fusion rule	Correctly calibrated estimates whose blind regions, residuals, and unresolved state components are explicit
Particle-physics forbidden and excluded regions	Experimental application	Separate definitions for theoretical prohibition, detector inaccessibility, and statistical exclusion	Reproducible limits whose dependence on model, acceptance, and confidence rule is explicit

P. Motivation and broader questions

P.1 Observation from within a system

An embedded observer uses its body, senses, instruments, memory, language, and inference rules to form a description of the world. Can it also determine how those means of representation condition its experience? Its evidence for that question must come through the same means of observation.

Philosophy offers several formulations of this question. Plato distinguishes appearances from what produces them. Aristotle distinguishes kinds of cause (Plato, 1992; Aristotle, 1984). Husserl examines how the structure of experience enters scientific descriptions (Husserl, 1970). Leibniz relates perspective to sufficient reason and the identity of indiscernibles (Leibniz, 1989; Leibniz and Clarke, 2000). Peirce connects the meaning of a concept to its conceivable practical consequences (Peirce, 1877, 1878). Whitehead emphasizes processes and relations (Whitehead, 1978).

The conjecture about self-observation is

> *No sufficiently expressive representational system can completely justify or perceive the boundary that generates its own representational capacity.*

The conjecture concerns an observer that must account for its own access. Formal self-reference has related limits in logic and semantics (Cantor, 1891; Russell, 1908; Gödel, 1931; Tarski, 1936;

Lawvere, 1969; Hofstadter, 1979, 2007). Kant distinguishes the conditions of experience from objects considered independently of those conditions (Kant, 1998). Second-order cybernetics includes the observer in the system it describes (Ashby, 1956; von Foerster, 1981; Maturana and Varela, 1980). Work on generalized observers and self-referential living systems asks how to formalize that relation (Elshatlawy et al., 2025; Abramsky et al., 2025). A general result for embedded physical observers remains open.

P.2 Recovery through a specified interface

A restricted version of the question has an exact answer. Given a source space and a fixed observation map, which differences between source states can an observation distinguish?

Let X be a source state space, let O be an accessible observation space, and let $\pi : X \to O$ be the observation map. When

$$\pi(x_1) = \pi(x_2) \qquad \text{for} \qquad x_1 \neq x_2,$$

the shared reading identifies a fiber containing both states. A function of that reading can return at most one of them as its answer.

The reconstruction theorem assumes a specified interface. A hidden extension shows that the same outputs can arise from different latent models. Whether a particular physical system must have a noninjective self-interface remains a separate question.

P.3 Competing models and the stopping rule

Several hypotheses can account for an observer who experiences a limit on its access.

(a) one foundational observer that generates a contained reality and relinquishes access to its own control;
(b) many genuinely independent observers within a shared physical world;
(c) a physical universe that produces bounded biological observers;
(d) an information-based reality without a singular foundational observer; and
(e) a cognitive model through which an embedded mind organizes severe uncertainty and self-reference.

If these models have the same possible reports, those reports alone cannot determine which model produced them.

A comparison requires a difference in predictions.

> *Treat the single-observer model, intentional forgetting, revelation, and every external-source model as metaphysical hypotheses unless they produce a testable prediction that distinguishes them from competing models, including through different outcome probabilities.*

Peirce and Popper relate inquiry to practical consequences and the possibility of failure (Peirce, 1877; Popper, 1959). Preregistered tests of consciousness theories specify outcomes that distinguish the competing predictions (Cogitate Consortium et al., 2025; Robinson et al., 2025).

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